

When Does Ability Grouping Improve Learning?

Evidence from Two Experiments in Kenya and Liberia*

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Abstract

When within-grade achievement variance is wide, schools can use a diagnostic assessment to reassign students across adjacent grades into ability-based instructional tracks. This cross-grade grouping can sharpen instructional targeting, but it also reorganizes classrooms—changing their size, peer composition, and age-grade structure. We study two randomized evaluations of cross-grade reading groups in Kenya and Liberia. Neither produces a positive average effect, but for different reasons. In Liberia, the baseline assessment explains less than 7% of endline variation; regrouping based on it does reduce average mismatch to track targets, but the assignment gains are modest and are swamped by the costs of reorganization—class sizes increase by a third. In Kenya, the baseline explains over 50% and regrouping substantially compresses within-class achievement dispersion, yet mean scores do not rise. Exploiting the fact that scripted curricula fix instructional content by track, we find evidence that adverse changes in peer composition offset the targeting gains. The contrast reveals a general tension in assignment under imperfect information: a diagnostic that improves on the incumbent rule along one dimension can worsen outcomes if acting on it disrupts the organizational structure—class sizes, peer groups, age cohorts—in which instruction takes place.

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1 Introduction

Schools must assign heterogeneous students to a limited number of instructional environments. The standard rule is grade, which groups students by age and prior progression through the curriculum. Grade is simple to administer, and it carries useful information about curricular exposure and developmental stage. Yet grade can be poorly aligned with actual ability when students within the same grade differ sharply in achievement and students in adjacent grades overlap substantially in skill.

The mismatch between grade and ability is particularly severe in many developing-country school systems. Within-grade achievement variance is wide, automatic promotion policies leave students far behind grade-level expectations, and multi-grade classrooms are common in small or under-resourced schools (Banerjee et al., 2007; Muralidharan et al., 2019). In these settings, a single teacher may face a classroom in which some students cannot recognize letters while others read fluently—a range that makes grade-level instruction poorly suited to most of the class. Cross-grade ability grouping is an appealing response. It requires no new teachers, materials, or technology—only a reassignment of existing students to existing classrooms based on a diagnostic assessment. And in schools with a single class per grade, within-grade tracking is not feasible: there are too few students per grade to form separate instructional groups without pooling across grades. Cross-grade grouping is therefore the natural organizational form when schools want to teach at assessed level rather than at enrolled grade. Recent evidence that aligning instruction to students’ demonstrated level can improve learning reinforces the case for trying (Banerjee et al., 2017; Duflo et al., 2020).

The underlying problem, though, is one of assignment under noisy information. Grade-based assignment and score-based assignment are both imperfect rules for instructional placement. Grade is coarse on measured achievement; a diagnostic score captures current performance more directly but may omit other determinants of successful placement—age, curricular exposure, classroom manageability. And regrouping changes the classroom itself: it alters peer composition, can increase or decrease class sizes, and may affect students’ relative standing within the group (Betts, 2011). A reclassification that improves targeting on paper may still fail if the resulting classrooms become less productive.

We study this question using two randomized evaluations of cross-grade reading groups in

schools operated by NewGlobe Education, a large school network that uses scripted lesson plans delivered via digital tablets. In 2018, approximately 90 Liberian public schools managed by NewGlobe were randomized to regroup students in Grades 1–4 into two reading tracks based on a baseline reading assessment. In the same year, 50 out of 300 NewGlobe academies in Kenya were randomized to regroup students in Grades 1–2 into two reading groups based on a combined literacy and English test. In both cases, regrouping applied to reading instruction only: students were reassigned to cross-grade reading classrooms for a daily instructional block, then returned to their enrolled grade for all other subjects. Because NewGlobe uses scripted curricula, instructional level in each track was determined by the track designation rather than by teacher discretion. This design is more demanding than within-grade tracking, which splits an existing grade cohort while preserving its age-grade structure. Cross-grade grouping uses the diagnostic rather than grade as the organizing rule for instruction—a change that can correct more severe instructional mismatch when learning levels overlap across grades, but that also removes students from their age cohort and reorganizes classroom composition more fundamentally.

Neither experiment produces a positive average treatment effect. The Liberia estimate is negative (-0.21 SD) and the Kenya estimate is a precisely estimated zero ($+0.02$ SD, $SE = 0.05$), ruling out effects as large as the $+0.18$ SD found by Duflo et al. (2011) for within-grade tracking. The two experiments differ sharply in the quality of the diagnostic signal and the severity of classroom reorganization, and the comparison reveals why each null arises.

In Liberia, the baseline assessment is a very weak predictor of endline achievement, explaining less than 7% of endline variation. Regrouping based on it does reduce average mismatch between students and track instructional targets, but it barely compresses within-class achievement dispersion—the sorted classrooms remain nearly as heterogeneous as the unsorted ones. Meanwhile, regrouping increases class sizes by roughly 13 students. The harm falls on the upper track (-0.34 SD, $p < 0.05$) and extends across the ability distribution rather than clustering near the assignment cutoff—a pattern more consistent with a deterioration of the upper-track classroom as a whole than with marginal misclassification of students near the threshold. Treated schools follow deterministic score cutoffs with zero observed assignment error; the modest assignment gains are real, but they are swamped by the costs of classroom reorganization.

In Kenya, the baseline is far more predictive, explaining over 50% of endline variation, and

regrouping does compress within-class dispersion. Yet mean achievement does not rise. The Kenya setting allows us to ask why. Because scripted curricula fix instructional content by track, two students with the same baseline score assigned to the same track receive the same instruction regardless of who else is in their classroom. This separates the peer-composition channel from instructional targeting—a distinction that is normally confounded in tracking studies. We find that higher peer ability *reduces* own achievement, consistent with students performing worse when regrouping lowers their standing relative to classmates. The sorting rule works as intended in Kenya; the problem is that the classrooms it creates are less productive than the ones it replaces.

A simple model of score-based grouping organizes these findings. It decomposes the effect of replacing grade-based with score-based assignment into four channels—assignment gains from reducing instructional mismatch, changes in productive instructional time, peer-composition effects, and relative-rank effects—and delivers a threshold condition: regrouping raises achievement only when the informational gain from the diagnostic exceeds the costs of classroom reorganization.

The paper makes three contributions. First, we provide the first experimental evidence on cross-grade ability grouping—what the early American literature called the Joplin Plan (Floyd, 1954).¹ Most meta-analyses of within-grade tracking find small or null average effects (Slavin, 1987, 1990; Kulik and Kulik, 1987; Terrin and Triventi, 2022), with heterogeneous results across tracks (Burke and Sass, 2013; Hoxby and Weingarth, 2005; Antecol et al., 2016; Garlick, 2018; de Roux and Riehl, 2022). Slavin (1987) judged the early Joplin Plan studies promising, but they were small and none used random assignment. In developing countries, well-identified evidence is scarce. The benchmark study is Duflo et al. (2011), who randomly assigned Kenyan primary schools to track first-graders into two within-grade sections and found average gains of +0.18 SD, with effects across the baseline ability distribution. Cross-grade grouping is a more demanding design. As we show in Section ?? (Remark 1), within-grade tracking holds grade fixed and asks the diagnostic only to rank students within a single grade’s ability distribution; it preserves age-grade structure and keeps within-classroom grade dispersion at zero by construction. Cross-grade grouping pools students across grades and requires the diagnostic to produce commensurable rankings across grade-specific distributions that may be shrunk at very different rates. It also mechanically increases grade

¹There is also a large body of research on between-school ability grouping, mainly from Europe where tracking between academic and vocational tracks is prevalent (Fu and Mehta, 2018; Lavrijssen and Nicaise, 2015). We are interested in within-school ability grouping.

mixing and can substantially change class sizes. Our experiments show that these additional costs can dominate even when the sorting rule succeeds on its own terms.

Second, the two-experiment design allows us to explain *why* ability grouping fails in each setting, not just report that it does. The tracking literature has documented considerable heterogeneity in effects across contexts (Betts, 2011; Card and Giuliano, 2016; Vardardottir, 2013) but has had difficulty explaining its sources because each study confounds multiple channels—instructional targeting, peer composition, class size, relative standing—that change simultaneously when students are regrouped. Our two experiments share the same institutional structure and implementation protocol but differ sharply in signal quality and classroom reallocation. In Kenya, because scripted curricula fix instructional content by track, the design creates leverage to examine the peer-composition channel separately from instructional targeting—a distinction that is normally confounded. Tracking also changes classroom dispersion, an input that studies of classroom production often overlook in favor of mean peer achievement (Rjosk, 2022; Vigdor and Nechyba, 2007; Duflo et al., 2011).

Third, we develop a framework for evaluating assignment rules that goes beyond the question of whether a diagnostic predicts achievement. The Teaching at the Right Level literature asks whether instruction targeted to assessed level raises scores (Banerjee et al., 2017; Duflo et al., 2020); proponents of tracking argue that reducing within-class dispersion allows teachers to target instruction more effectively (Duflo et al., 2011). Our framework asks a different question: whether a score-based assignment rule improves on the incumbent rule—grade—after accounting for the organizational costs of acting on it. Tracking changes not only the level of instruction but also the composition and size of the classroom, and the literature on mixed-grade classrooms suggests that the sign of these effects is context-dependent (Sims, 2008; Leuven and Rønning, 2011). The model delivers a threshold condition that depends jointly on signal quality and classroom reorganization, and the two experiments land on opposite sides of the margins that matter.

The paper proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the experiments and data. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses robustness. Section 7 concludes.

2 Theoretical Framework

We model cross-grade ability grouping as a classification problem under a noisy signal. Students differ in latent ability, but schools observe only an imperfect diagnostic. A score-based reassignment rule can improve achievement by placing students in classrooms better matched to their level, but it also reorganizes the classroom: changing class sizes, mixing students from different developmental stages, shifting peer composition, and altering students' relative standing.

2.1 Environment and Assignment

Student i in grade $g(i) \in \mathcal{G}$ has latent ability

$$\theta_i = \mu_{g(i)} + \nu_i, \tag{1}$$

where $\mu_{g(i)}$ is the grade-specific mean and ν_i captures within-grade heterogeneity, with $\mathbb{E}[\nu_i] = 0$ and $\text{Var}(\nu_i \mid g(i)) = \sigma_{\nu,g}^2$. The school does not observe θ_i directly. It observes a noisy diagnostic score

$$s_i = \theta_i + u_i, \tag{2}$$

with $\mathbb{E}[u_i \mid \theta_i, g(i)] = 0$ and $\text{Var}(u_i \mid g(i)) = \sigma_{u,g}^2$. The school's problem is analogous to an employer deciding whether to use a test score in addition to an observable credential (here, grade) when assigning workers to tasks (Autor and Scarborough, 2008). Grade already provides some information about ability; the diagnostic is valuable only if it moves the school meaningfully beyond what grade already reveals. When the test is noisy relative to the within-grade ability distribution, the posterior $\mathbb{E}[\theta_i \mid s_i, g(i)]$ is pulled back toward the grade mean, and reassignment based on the score is little better than leaving students in their enrolled grade.

The posterior expectation of ability given the signal and grade is

$$a_i \equiv \mathbb{E}[\theta_i \mid s_i, g(i)] = \mu_{g(i)} + \rho_g(s_i - \mu_{g(i)}), \tag{3}$$

where

$$\rho_g \equiv \frac{\sigma_{\nu,g}^2}{\sigma_{\nu,g}^2 + \sigma_{u,g}^2} \tag{4}$$

is the signal quality of the diagnostic. When ρ_g is high, the posterior tracks the raw score closely and the diagnostic carries substantial information beyond grade; when ρ_g is low, the posterior is pulled toward the grade mean and reassignment based on the test is only a weak improvement over grade assignment. The parameter ρ_g is not directly observed. Our empirical proxy is R_g^2 , the within-grade R^2 from regressing endline scores on baseline scores in the control group. Under the model’s Gaussian structure, $R_g^2 = \rho_g \cdot \sigma_{\nu,g}^2 / (\sigma_{\nu,g}^2 + \sigma_{\varepsilon,g}^2)$, where $\sigma_{\varepsilon,g}^2$ is endline noise; because endline noise attenuates the correlation, $R_g^2 < \rho_g$, so the empirical measure is a conservative proxy for the model’s shrinkage weight. We refer to R_g^2 as *baseline predictive power*. It blends test precision, achievement persistence, and baseline-endline construct alignment into a single measure of how useful the diagnostic is for assignment—whatever the source of a low R_g^2 , a signal with low predictive power provides less useful information for reassignment.

Under the status quo regime (G), students remain in grade-based classrooms and receive grade-level instruction $I_i^G = I(g(i))$. Under score-based grouping (T), the school assigns students to a low or high instructional track using the posterior signal:

$$k_i^T = \begin{cases} L & \text{if } a_i < c, \\ H & \text{if } a_i \geq c, \end{cases} \quad I_i^T = \begin{cases} I_L & \text{if } k_i^T = L, \\ I_H & \text{if } k_i^T = H. \end{cases} \quad (5)$$

Under scripted curricula—where lesson plans are standardized and teacher discretion over content is limited—instructional level is fixed by track designation rather than adjusted by the teacher. This means that the instructional content a student receives depends only on which track she is assigned to, not on who else ends up in her classroom. Both experiments we study operate under scripted instruction (Section 3). The assignment decision—high or low track—and the classroom-reorganization consequences—peer quality, class size—are therefore institutionally distinct, even though they are jointly determined by the same score rule.

2.2 Achievement

Endline achievement under assignment regime $a \in \{G, T\}$ is

$$Y_i(a) = \theta_i + \tau_{c(i,a)} \left[\alpha - \lambda(\theta_i - I_i(a))^2 + \pi \bar{\theta}_{-i,c(i,a)} + \eta r_{i,c(i,a)} \right] + \varepsilon_i(a). \quad (6)$$

Baseline ability θ_i persists into the endline: a student who learns nothing during the intervention still retains her initial achievement level. The bracketed term is the value added by instruction, which depends on how well instruction is matched to the student’s level, who her classmates are, and how much classroom time is productively used. This term is scaled by τ , so all instructional channels are attenuated when the classroom becomes harder to manage.

The term $(\theta_i - I_i(a))^2$ is *instructional mismatch*: a student whose ability is far from the instructional target of her classroom learns less than one whose ability is close to it, regardless of whether the instruction is too advanced or too basic. The quadratic form implies that the cost of mismatch is convex—small deviations matter little, but large deviations are increasingly costly. Under grade-based assignment, mismatch arises because students within the same grade differ in ability but receive the same instruction. Under score-based grouping, the diagnostic can reduce mismatch by sorting students into tracks whose instructional targets are closer to their ability, but only if the diagnostic is informative enough for the sorting to be accurate.

The term $\bar{\theta}_{-i,c(i,a)}$ captures the level of peer ability in student i ’s classroom, while $r_{i,c(i,a)} \equiv \text{rank}_i / N_{c(i,a)}$ denotes student i ’s percentile rank within that classroom. We normalize by class size so that rank is comparable across classrooms of different sizes—an important consideration here, since tracking changes class sizes. Both terms are needed because regrouping can place a student among stronger peers while simultaneously lowering that student’s relative standing, so the net social effect of tracking is not signed *a priori*. The sign of π is theoretically ambiguous—positive under peer learning, negative if lower-ability peers generate disruption. The coefficient η captures relative-standing effects: percentile rank affects achievement independently of the level of surrounding peers (Murphy and Weinhardt, 2020; Elsner et al., 2021), consistent with the “big-fish-little-pond” effect in educational psychology (Hoxby and Weingarth, 2005).

The object $\tau_{c(i,a)} \in (0, 1]$ is the fraction of classroom time that is productively engaged. Following the logic of Lazear (2001), productive time is a classroom-level public good: when the classroom becomes harder to manage, all students lose instructional time. We let

$$\tau_c = \tau(N_c, V_c^g), \quad \frac{\partial \tau}{\partial N_c} < 0, \quad \frac{\partial \tau}{\partial V_c^g} < 0, \quad (7)$$

where N_c is class size and $V_c^g \equiv \text{Var}_{i \in c}(g(i))$ is within-classroom grade dispersion. The class-size

channel is standard: larger classrooms reduce the share of time available per student. The grade-dispersion channel is specific to cross-grade grouping. A diagnostic sorts students on one measured dimension (reading level), but students from different grades differ on unmeasured dimensions—behavioral maturity, attention span, familiarity with classroom routines—that affect how easily the classroom functions as a unit. When regrouping pools adjacent grades, each student’s probability of disrupting is increasing in the distance between that student’s developmental stage and the classroom’s modal stage; the aggregate disruption rate is therefore increasing in within-class grade dispersion, which rises mechanically in treatment classrooms.

Because τ multiplies the entire bracketed term, a fall in productive time attenuates all other channels: if regrouping creates a classroom that is too large or too heterogeneous in developmental stage for the teacher to manage, the returns to better instructional match, stronger peers, or higher rank are all diminished. Correct placement does not help much in a classroom where effective instruction is not occurring.

2.3 Treatment Effect Decomposition

Define student i ’s *assignment gain* as

$$B_i \equiv (\theta_i - I_i^G)^2 - (\theta_i - I_i^T)^2. \quad (8)$$

A positive B_i means that score-based grouping moves student i closer to the instructional level that fits her ability. Let

$$A_i(a) \equiv \alpha - \lambda(\theta_i - I_i(a))^2 + \pi \bar{\theta}_{-i,c(i,a)} + \eta r_{i,c(i,a)} \quad (9)$$

collect the within-brackets terms. The treatment effect for student i separates into two parts: changes in the quality of instruction holding productive time fixed, and changes in productive time holding instruction quality at its treatment level:

$$\Delta Y_i \equiv Y_i(T) - Y_i(G) = \tau_{c(i,G)} [A_i(T) - A_i(G)] + [\tau_{c(i,T)} - \tau_{c(i,G)}] A_i(T) + \tilde{\varepsilon}_i, \quad (10)$$

where $\tilde{\varepsilon}_i = \varepsilon_i(T) - \varepsilon_i(G)$. Expanding gives

$$\Delta Y_i = \underbrace{\lambda \tau_{c(i,G)} B_i}_{\text{assignment gain}} + \underbrace{\pi \tau_{c(i,G)} \Delta \bar{\theta}_i}_{\text{peer level}} + \underbrace{\eta \tau_{c(i,G)} \Delta r_i}_{\text{rank}} + \underbrace{[\tau_{c(i,T)} - \tau_{c(i,G)}] A_i(T)}_{\text{productive time}} + \tilde{\varepsilon}_i, \quad (11)$$

where $\Delta \bar{\theta}_i \equiv \bar{\theta}_{-i,c(i,T)} - \bar{\theta}_{-i,c(i,G)}$ and $\Delta r_i \equiv r_{i,c(i,T)} - r_{i,c(i,G)}$.

Score-based grouping therefore affects achievement through four channels: it can improve instructional match (B_i), change peer composition ($\Delta \bar{\theta}_i$), change relative rank (Δr_i), and alter the amount of productive instructional time available ($\Delta \tau$). Averaging over students yields

$$\text{ATE} = \lambda \mathbb{E}[\tau_{c(i,G)} B_i] + \pi \mathbb{E}[\tau_{c(i,G)} \Delta \bar{\theta}_i] + \eta \mathbb{E}[\tau_{c(i,G)} \Delta r_i] + \mathbb{E}[(\tau_{c(i,T)} - \tau_{c(i,G)}) A_i(T)]. \quad (12)$$

2.4 Signal Quality and the Value of Grouping

Under the Gaussian structure, the posterior a_i concentrates around θ_i as ρ_g rises, so track assignments become more accurate and the reduction in mismatch grows.

Proposition 1. *Under Gaussian ν_i and u_i with a monotone cutoff rule:*

- (i) *The expected assignment gain $\mathbb{E}[B_i \mid \rho_g]$ is strictly increasing in ρ_g on $(0, 1)$, with $\mathbb{E}[B_i \mid \rho_g = 0] = 0$ and $\mathbb{E}[B_i \mid \rho_g = 1] = \mathbb{E}[(\theta_i - I_i^G)^2 - (\theta_i - I_i^{T*})^2]$, where I_i^{T*} is the oracle-optimal track target under perfect observation of θ_i .*
- (ii) *If $\text{ATE}(\rho_g = 1) > 0$ and $\text{ATE}(\rho_g)$ is continuous in ρ_g , then since $\text{ATE}(0) \leq 0$, there exists at least one $\rho_g^* \in (0, 1)$ such that $\text{ATE}(\rho_g^*) = 0$.*

Proof sketch. Part (i): When $\rho_g = 0$, the posterior $a_i = \mu_{g(i)}$ for all students in the same grade: the test provides no information beyond grade, so track assignment is determined entirely by grade and $\mathbb{E}[B_i] = 0$. As ρ_g rises, the posterior concentrates around θ_i , a strictly larger fraction of students are assigned to the track that minimizes their mismatch, and $\mathbb{E}[B_i]$ rises continuously. At $\rho_g = 1$, the posterior equals true ability and assignment is oracle-optimal given the available tracks, yielding the maximal mismatch reduction. Since $\tau_{c(i,G)}$ does not depend on ρ_g (it is determined by the status quo classroom), $\mathbb{E}[\tau_{c(i,G)} B_i]$ inherits the monotonicity. Part (ii): At $\rho_g = 0$, the assignment-gain term is zero while the reorganization terms—peer composition, rank, productive time—can

be negative (regrouping still moves students across classrooms without informational gain). So $\text{ATE}(0) \leq 0$. If $\text{ATE}(1) > 0$, continuity gives at least one crossing. $\square$

The reorganization terms— $\mathbb{E}[\tau_{c(i,G)}\Delta\bar{\theta}_i]$, $\mathbb{E}[\tau_{c(i,G)}\Delta r_i]$, and $\mathbb{E}[(\tau_T - \tau_G)A_i(T)]$ —are themselves functions of ρ_g through the assignment, so the threshold need not be unique. When the signal is poor, near-random reassignment simultaneously fails to reduce mismatch, scrambles peer composition, and—in settings where regrouping increases class sizes or grade mixing—reduces productive time. All four terms in equation (12) can be negative simultaneously. The production function parameters $(\lambda, \pi, \eta, \tau)$ need not be the same across settings. The model asks whether the assignment gain outweighs the reorganization costs; the answer can differ across settings because signal quality, class sizes, and composition shifts differ.

Remark 1 (Within-grade vs. cross-grade tracking). *The decomposition in equation (12) applies to any score-based reassignment, but the magnitudes of the terms differ systematically between within-grade tracking (splitting a single grade into tracks) and cross-grade tracking (pooling adjacent grades and re-sorting).*

Productive time. Under within-grade tracking, all students in a classroom share the same grade, so $V_c^g = 0$ by construction. Under cross-grade tracking, treatment classrooms pool students from different grades, so $V_c^g > 0$. Since $\partial\tau/\partial V_c^g < 0$, cross-grade tracking incurs a productive-time cost that within-grade tracking avoids entirely. Class sizes also differ: within-grade tracking splits an existing grade-level classroom roughly in half ($N_c^{WG} \approx N_g/2$), while cross-grade tracking determines class sizes by the ability distribution across the pooled grades, which can substantially increase N_c .

Assignment gain. Under within-grade tracking with a grade-specific cutoff c_g , student i 's assignment depends on $a_i - c_g = \rho_{g(i)}(s_i - \mu_{g(i)}) - (c_g - \mu_{g(i)})$, which involves only the signal quality $\rho_{g(i)}$ of her own grade. Under cross-grade tracking with a common cutoff c , student i 's assignment depends on

$$a_i - c = \mu_{g(i)} + \rho_{g(i)}(s_i - \mu_{g(i)}) - c. \quad (13)$$

Whether student i in grade g is assigned above or below a student j in grade g' now depends on both ρ_g and $\rho_{g'}$. When signal quality varies across grades, posteriors are shrunk at different rates: a student in a high- ρ grade is ranked largely on ability, while a student in a low- ρ grade is ranked largely on her grade mean. The cross-grade ranking conflates these differently-informative signals,

distorting assignment in a way that within-grade tracking does not face.

3 Experimental Design and Data

3.1 Setting

Both experiments take place in schools operated by NewGlobe Education. NewGlobe operates large networks of low-cost schools using a standardized model: scripted lesson plans delivered through digital tablets, with teacher compliance monitored by school managers. Because lesson content is scripted, instructional level is determined by track designation rather than by teacher discretion—a feature that plays a central role in the peer-effects identification strategy (Section 4.3).

3.2 Experiment 1: Kenya Year 1 (2018)

In August 2018, 50 out of approximately 300 NewGlobe academies in Kenya were randomized to receive the reading group intervention. In treated schools, students in Grades 1 and 2 were assigned to one of two reading groups (“Blue” and “Purple”) based on a combined literacy and English baseline test administered in June 2018. Grade 1 students scoring above 40 points were assigned to the Purple (upper) group; Grade 2 students above 35 points. Treatment was stratified by constituency. Students received 90 minutes of instruction with their reading group daily for Term 3 (approximately 3 months).

3.3 Experiment 2: Liberia (2018)

Approximately 90 public schools in Liberia, managed by NewGlobe under the Liberia Education Advancement Program, were randomized into treatment and control. In treatment schools, students in Grades 1–4 were regrouped: Grades 1 and 2 were combined and split into two tracks, as were Grades 3 and 4. The diagnostic test was a baseline reading assessment; the cutoff for the upper track was 23 points for the Grade 1–2 pool and 14 points for the Grade 3–4 pool. Treatment was stratified by grade group and treatment probability.

3.4 Data

Baseline data consist of literacy and English language test scores administered before the intervention in each experiment. Endline data consist of literacy assessments administered at the end of the school year. In Kenya, students in different grades took grade-specific tests; we standardize all scores to have mean zero and unit standard deviation in the control group within each grade. Tables A2 and A3 in the appendix document the sample construction.

Table 1 reports summary statistics for the Liberia sample, and Table 2 reports summary statistics for Kenya Year 1. Figure 2 plots the baseline score distributions by grade and treatment status; the substantial overlap across grades confirms the scope for cross-grade regrouping.

3.5 Balance and Attrition

Tables 3 and 4 report covariate balance across treatment and control schools. Pre-treatment characteristics—baseline scores, grade composition, school size—are jointly balanced in both experiments, consistent with successful randomization.

Table ?? reports attrition rates by treatment status, experiment, and baseline-ability quartile. Overall attrition is balanced in Liberia. In Kenya Year 1, attrition is modestly lower in treated schools, but the difference is small and does not vary systematically with baseline ability. To bound the potential impact of differential attrition, Table ?? reports ? trimming bounds. The bounded intervals do not materially change the Liberia inference and still include zero for Kenya Year 1, so selective attrition cannot explain the main results.

4 Empirical Strategy

4.1 Intent-to-Treat Effects

We estimate the overall treatment effect using:

$$Y_i = \alpha + \tau T_j + \gamma \hat{\theta}_i^{EB} + \phi_s + \varepsilon_i \quad (14)$$

where Y_i is the standardized endline score, T_j is school-level treatment assignment, ϕ_s are strata fixed effects, and standard errors are clustered at the school-grade-group level. Because treatment was randomized at the school level, OLS identifies the average intent-to-treat effect τ . We also estimate effects by grade group and by track assignment (upper vs. lower), interacting treatment with the relevant indicators.

The baseline control $\hat{\theta}_i^{EB}$ is an empirical Bayes ability measure that shrinks each student’s raw score toward the grade mean by the grade-specific signal quality:

$$\hat{\theta}_i^{EB} = \bar{s}_{g(i)} + \hat{\rho}_{g(i)} (s_i - \bar{s}_{g(i)}), \quad (15)$$

where $\hat{\rho}_{g(i)} \equiv \sqrt{R_{g(i)}^2}$ and R_g^2 is the within-grade baseline-endline R^2 in the control group (Section ??). Because $R_g^2 < \rho_g$, the EB measure shrinks more aggressively than the model’s posterior, making the resulting ability estimates conservative.² Beyond its role as a baseline control, $\hat{\theta}^{EB}$ is used to construct the mismatch measure $\text{Misfit}_i = (\hat{\theta}_i^{EB} - \hat{I}_{k(i)})^2$, where $\hat{I}_{k(i)}$ is the estimated target ability level of track $k(i)$. Grade-specific shrinkage ensures that misfit downweights score variation in grades where the test is uninformative, so that the measure reflects genuine ability dispersion rather than measurement error.

4.2 Effect on Within-Class Dispersion

Before asking whether tracking raises achievement, we ask a prior question: does it actually compress within-class ability heterogeneity? The answer is not obvious. Tracking sorts students on a noisy signal, so the realized compression depends on how informative the diagnostic is. If signal quality is low, the reassignment is close to random and within-class dispersion falls little.

We measure dispersion using baseline scores rather than endline scores. Baseline scores are pre-determined, so within-class baseline dispersion is a purely mechanical function of who the sorting rule places together—it is not affected by any treatment-induced changes in achievement. Using endline scores would conflate the sorting effect with treatment effects on the distribution of

²When $\hat{\rho}_g$ is constant across grades, $\hat{\theta}^{EB}$ is an affine transformation of s_i and the mismatch measure is a rescaled version of raw-score misfit. Grade-specific shrinkage matters here because R_g^2 ranges from ≈ 0.03 (Liberia Grade 3) to ≈ 0.56 (Kenya Grade 2). In the ITT specification, $\hat{\theta}_i^{EB}$ enters linearly as a baseline covariate; with grade fixed effects already in the model, grade-specific shrinkage is equivalent to interacting baseline scores with grade.

outcomes. We estimate:

$$|s_i - \bar{s}_{k(i)}| = \alpha + \tau T_j + \gamma \hat{\theta}_i^{EB} + \phi_s + \varepsilon_i \quad (16)$$

where $|s_i - \bar{s}_{k(i)}|$ is student i 's absolute deviation from the mean baseline score in her reading classroom $k(i)$. A negative τ indicates that tracking compresses within-class heterogeneity. Figure 3 plots the classroom-level standard deviation of baseline scores in treatment and control classrooms directly.

4.3 Isolating Peer Effects

When a school is randomized into tracking, everything about the classroom changes at once: peer composition, instructional match, class size, grade mixing. A naive regression of outcomes on peer ability conflates the direct effect of peers with all of these correlated changes. To make progress, we need variation in peer composition that is not driven by variation in other classroom characteristics.

The experiment provides such variation. Consider two students in different treated schools who have the same baseline score and are therefore assigned to the same track. Their instructional targets are identical, but their classmates differ because the composition of each school differs. One student may end up in a reading group where the other students in her track happen to be relatively strong; another may end up with weaker peers. This variation in peer composition is not chosen by the student or the school—it is a mechanical consequence of which other students happen to attend the same school, combined with the exogenous treatment assignment that determines whether and how students are regrouped.

Borusyak and Hull (2023) formalize the conditions under which this kind of variation identifies a causal peer effect. Let $P_i = (1/n_k) \sum_{j \neq i} s_j$ denote the mean baseline score of student i 's classmates. P_i is a known, deterministic function of two objects: the treatment assignment T_j and the vector of predetermined baseline scores in the school. The identification strategy conditions on what we would *expect* P_i to be given student i 's own score and treatment status:

$$\mu_P(s_i, T_j) = \mathbb{E}[P_i \mid s_i, T_j]. \quad (17)$$

Once we condition on this expected value, the remaining variation in P_i —the deviation $\tilde{P}_i =$

$P_i - \mu_P(s_i, T_j)$ —reflects only the idiosyncratic composition of the particular school, not the student’s own characteristics or treatment status.

Three assumptions are required:

Assumption 1 (Random Assignment). $T_j \perp (\theta_i, u_i, \varepsilon_i)$ for all $i \in j$.

Assumption 2 (Conditional Independence). $\{s_k\}_{k \in j, k \neq i} \perp (\theta_i, \varepsilon_i) \mid s_i, T_j$.

Assumption 3 (Scripted Instruction). *The instructional level I_{kg} in each track k is a deterministic function of the track designation and is not adjusted by the teacher in response to realized classroom composition.*

Assumption 1 holds by design. Assumption 2 requires that, conditional on a student’s own score and treatment status, the scores of her schoolmates are uninformative about her unobserved determinants of achievement—a within-school version of the standard peer-effects exclusion restriction. Under these two assumptions, the recentered instrument $\tilde{P}_i$ satisfies $\mathbb{E}[\tilde{P}_i \cdot \varepsilon_i \mid s_i, T_j] = 0$.

We implement this by including baseline-decile $\times$ treatment fixed effects, which approximate $\mu_P(s_i, T_j)$:

$$Y_i = \beta_P P_i + \sum_{d,t} \delta_{dt} \cdot \mathbb{1}\{D_i = d, T_j = t\} + \phi_s + \varepsilon_i. \quad (18)$$

The decile $\times$ treatment cells absorb the expected peer composition for each type of student up to within-decile variation in s_i , so β_P is identified from deviations of actual peer composition around that expectation. The conditioning is approximate: deciles discretize the continuous score, so residual within-decile variation in s_i is not absorbed. We probe sensitivity to finer bins (ventiles) in Section 6.

Assumption 3 plays a distinct role. In most tracking settings, instructional match and peer composition move together and cannot be separated: a student placed in a higher track gets both better-matched instruction and different peers. With scripted curricula, instructional level is fixed by track designation. The mismatch measure $\hat{M}_i = (s_i - I_{D_i})^2$ is therefore approximately a deterministic function of (s_i, T_j) : it is absorbed by decile $\times$ treatment fixed effects up to within-decile score variation. This approximately breaks the collinearity between peer composition and instructional match, so that β_P captures the peer-composition effect net of mismatch to the extent that the discretization is fine enough.

4.4 Decomposing the Treatment Effect

The model decomposes the ATE into four terms (equation 12): assignment gain, peer composition, relative rank, and productive time.

The return to instructional match λ is estimated from:

$$Y_i = \beta_0 - \lambda \cdot \text{Misfit}_i + \gamma_1 \hat{\theta}_i^{EB} + \gamma_2 (\hat{\theta}_i^{EB})^2 + \varepsilon_i \quad (19)$$

where $\text{Misfit}_i = (\hat{\theta}_i^{EB} - \hat{I}_{k(i)})^2$, the squared distance between a student’s predicted ability and the instructional target of the assigned track. The quadratic ability control is essential to avoid confounding the convex ability-outcome relationship with the mismatch penalty. Combined with the estimated treatment-induced change in track-target mismatch and classroom composition, this yields $\hat{\lambda} \hat{\mathbb{E}}[B_i]$.

The recentered peer-effects regression in Section 4.3 estimates $\hat{\beta}_P$, the effect of peer composition conditional on own ability and treatment status. This coefficient conflates peer-level effects (π) with rank effects (η), since peer mean and within-class rank are correlated in tracked classrooms. The big-fish-little-pond exercise reported in Section 5 provides suggestive evidence on the rank channel specifically.

Productive time is parameterized as

$$\tau_c = \exp(-\psi_N N_c - \psi_V V_c^g), \quad (20)$$

where N_c is class size and V_c^g is within-classroom grade dispersion. The exponential ensures $\tau_c \in (0, 1]$ and nests the reduced-form exercise of controlling for class size as a special case ($\psi_V = 0$). We also estimate the treatment effect on class size directly and examine how controlling for class size changes the ITT.

The Kenya data are rich enough to estimate the decomposition jointly. We stack the mean and quartile-specific ITT estimates as a vector of moments and choose $(\lambda, \pi + \eta, \psi_N, \psi_V)$ to minimize the distance between these moments and their model-implied counterparts, computed from the observed joint distribution of baseline scores, track assignments, peer composition, and classroom characteristics. Gaussian assumptions on ability and measurement error ($\nu_i \sim N(0, \sigma_{\nu,g}^2)$, $u_i \sim$

$N(0, \sigma_{u,g}^2)$) generate the posterior and the assignment rule; the exponential τ specification closes the model. Standard errors are obtained by bootstrap. Because peer-mean and rank move together in tracked classrooms, the estimator recovers a composite social-composition parameter $\zeta \equiv \pi + \eta$ rather than separate peer-level and rank effects.

In Liberia, the assignment-gain term has too little variation to identify λ separately: with $R_g^2 < 0.07$, the posterior is pulled so strongly toward the grade mean that the treatment-induced change in mismatch is small and nearly constant across students, leaving the estimator unable to distinguish mismatch effects from the intercept. We therefore do not estimate the production function in Liberia. Instead, we take the Kenya composition estimate $\hat{\zeta}$ as given and map out the regions of the (λ, ψ) parameter space where grouping would raise or lower achievement in the Liberia setting, with and without conservative assumptions about classroom-disruption penalties.

5 Results

5.1 Treatment Effects on Test Scores

Table 5 presents the main intent-to-treat estimates. In Liberia, tracking reduces endline scores by 0.205 SD, but the effect is not statistically significant at conventional levels ($p = 0.13$). In Kenya Year 1, the effect is essentially zero (+0.023 SD, $SE = 0.054$). The Kenya estimate is precise enough to rule out effects as large as the +0.18 SD found by Duflo et al. (2011); the minimum detectable effect at 80% power is 0.15 SD. The Liberia estimate is less precise (MDE = 0.38 SD), but the point estimate is negative, and the pattern beneath the average is informative.

Table 6 breaks the Liberia result by track. Upper-grade students (Grades 3–4) are more negatively affected (−0.358 SD, $p < 0.10$) than lower-grade students (−0.089 SD, insignificant), and students assigned to the upper track are harmed by an additional −0.33 SD relative to lower-track students, for a total upper-track effect of −0.342 SD ($p < 0.05$). In Kenya, the upper-lower split is uninformative: both track-specific effects are close to zero.

5.2 Signal Quality, Assignment, and Classroom Reallocation

Table 8 and Figure 6 report baseline predictive power. In Kenya Year 1, R_g^2 is 0.509 in Grade 1 and 0.559 in Grade 2. In Liberia, the corresponding values are only 0.034–0.065 across grades. Figure 6

makes the consequence visible: in Kenya, the empirical Bayes posterior remains responsive to the raw score, so the diagnostic carries real information beyond grade; in Liberia, the posterior is pulled so strongly toward the grade mean that the score-based assignment barely differs from assigning students by grade alone. Figure 4 shows the same contrast in raw baseline-endline comovement. This R^2 -based measure blends test precision, achievement persistence, and construct alignment, but the Kenya-Liberia contrast is robust to alternative measures: Table 9 shows that within-grade rank persistence from baseline to endline is much higher in Kenya (0.684) than in Liberia (0.242), and that the incremental predictive power of baseline scores over grade fixed effects is an order of magnitude larger in Kenya (0.522 versus 0.050).

A plausible explanation is that signal quality is not a property of the test instrument alone—it is endogenous to the institutional environment in which the test is administered. In Kenya, Bridge operated its own private-school network, building and running schools under a highly centralized model with control over curriculum, teacher training, and school operations. In Liberia, by contrast, Bridge participated in the Partnership Schools for Liberia reform by managing existing public schools that remained government-owned, free to students, non-selective, and staffed by civil-service teachers on the government payroll. In a weaker institutional environment, the same forces that make instruction less effective—less trained teachers, less structured classrooms, wider heterogeneity in what students have actually been exposed to—also make a one-time diagnostic less predictive. The test is administered under the same constraints that limit learning, and the students it measures have followed less standardized paths to the assessment. Signal quality and system quality are jointly determined, not independent.

Table 10 and Figure 7 confirm that treated schools in both experiments follow the published cutoffs exactly, with zero observed assignment error. In Liberia, the diagnostic is applied precisely, but to a signal that carries little information.

Table 11 and Figure 8 show what reassignment does to classrooms. Regrouping lowers track-target mismatch in both settings: the mean mismatch measure falls from 1.35 to 0.49 in Kenya and from 1.52 to 0.82 in Liberia. The classroom consequences, however, diverge. In Kenya, treated reading classes are smaller on average (20.9 to 18.0 students), though they become more mixed-grade. In Liberia, treated reading classes grow substantially (32.2 to 42.2 students) and become equally more mixed-grade. Figure 3 shows that the compression in within-class baseline-score

dispersion is also much stronger in Kenya than in Liberia.

5.3 Treatment Effect Heterogeneity

The concentration of harm in the Liberia upper track could reflect misclassification: students with scores just above the cutoff may truly belong in the lower track, and placing them in the upper track hurts them individually. Table 17 tests this by breaking upper-track students into within-track ability terciles. If misclassification were the mechanism, harm should be concentrated in the bottom tercile—students most likely to truly belong in the lower track. Instead, harm is largest for the middle (-0.48 SD, $p < 0.10$) and top (-0.45 SD, $p < 0.01$) terciles, while the bottom tercile effect is smaller and insignificant (-0.20 SD). Table 16 confirms the pattern from the other direction: treatment harm *increases* with distance from the assignment cutoff rather than concentrating near it (Figure 5).

This pattern is more consistent with a deterioration of the upper-track classroom as a whole. Table 11 shows that regrouping increases mean reading-class size in Liberia from 32.2 to 42.2 students while increasing mixed-grade exposure. Controlling for class size reduces the upper-track interaction from -0.33 to -0.23 SD, so congestion contributes but does not fully account for the harm. The number of comparisons grows as we cut by track, grade block, and distance from cutoff, and the cell sizes shrink accordingly, so these patterns should be read as suggestive.

5.4 Peer Effects

In Kenya, accurate sorting compresses within-class achievement dispersion (Figure 3) but does not raise mean scores. A natural candidate explanation is that regrouping changes peer composition in ways that offset the targeting gains. We estimate the recentered regression described in Section 4.3 to isolate the effect of peer composition on achievement.

Table 7 reports the estimates. In Liberia, peer ability has no detectable effect on achievement ($\hat{\beta}_P = -0.006$, $SE = 0.077$). In Kenya Year 1, a one-SD increase in peer ability *reduces* own achievement by 0.133 SD ($p < 0.05$). Scripted instruction means that instructional level is fixed by track designation, so the decile $\times$ treatment fixed effects absorb the mismatch channel; adding classroom-level controls does not attenuate the coefficient. The cross-site pattern is also consistent with the identification strategy: the coefficient is zero in Liberia, where signal quality is low and

treatment-induced variation in peer composition is small, and negative in Kenya, where signal quality is high and peer composition shifts substantially.

The coefficient should be interpreted with two qualifications. Because peer mean and within-class rank move together in tracked classrooms, $\hat{\beta}_P$ conflates peer-level effects with rank effects. The big-fish-little-pond pattern—the negative coefficient is larger for students whose relative standing drops most—is consistent with rank effects contributing, but cannot separate them cleanly. And the Liberia null is also consistent with measurement-error attenuation: given Liberia’s low signal quality, peer means are measured far less precisely, and a coefficient of Kenya magnitude would be hard to detect.

5.5 Kenya Accounting Exercise and Liberia Sensitivity

Table 12 reports the Kenya sufficient-statistics decomposition. The peer/rank composite is negative and statistically meaningful ($\hat{\zeta} = -0.125$, $SE = 0.055$), while the mismatch return is near zero and imprecise ($\hat{\lambda} = 0.003$, $SE = 0.017$). The decomposition implies a small negative peer/rank contribution and a near-zero assignment contribution, with a positive residual needed to match the overall ITT. The preferred minimum-distance estimator (Table 13) remains a useful specification check; Figure 9 shows that it tracks the mean and quartile ITT moments closely.

The parameters are not equally well identified. Bootstrap and profile exercises show that the mismatch penalty is more stable than the composition and disruption terms; the lower end of the bootstrap interval for λ still reaches zero. The decomposition establishes internal consistency—the Kenya null can be rationalized by a model in which assignment gains and reallocation costs roughly cancel—rather than pinning down each parameter tightly.

In Liberia, signal quality is too low for the same exercise: the assignment-gain term has too little variation to identify λ separately, so we instead map out the regions of the (λ, ψ) parameter space where grouping would raise or lower achievement, taking the Kenya composition estimate $\hat{\zeta}$ as given. Table 14 shows that the Liberia BH peer coefficient is near zero and that transported Kenya channels explain little of the Liberia ITT, leaving a large residual. Table 15 and Figures 10–11 report the sensitivity results. The pooled result masks heterogeneity across grade blocks: in Grades 1–2, the favorable region is essentially absent even under the baseline specification; in Grades 3–4, it persists under conservative assumptions. Bootstrap intervals for the beneficial-share

statistics are wide, and the social-composition term is transported from Kenya, so the maps should be read as sensitivity exercises conditional on a Kenya-based composition estimate rather than free-standing Liberia structural estimates.

6 Robustness

Table ?? reports the ITT estimates under alternative specifications—school-level clustering, wild bootstrap p -values, models without baseline controls, and permutation-based inference. The point estimates are stable across all specifications; the qualitative conclusions do not depend on the choice of standard-error adjustment or baseline-score control.

Tracking changes class size in opposite directions across the two experiments: treated classrooms shrink in Kenya (from 20.9 to 18.0 students) and grow substantially in Liberia (from 32.2 to 42.2). The Liberia increase reflects the mechanics of combining two grades’ worth of students into two tracks within existing school infrastructure. This raises the concern that the Liberia results are driven by overcrowding rather than by the sorting mechanism the model emphasizes. Table ?? reports the upper-track interaction with and without class-size controls. Controlling for class size attenuates the Liberia upper-track interaction from -0.33 to -0.23 SD: congestion contributes roughly 30% of the upper-track harm, but the majority of the effect remains after adjustment. In Kenya, controlling for class size does not change the peer-composition estimates.

The Kenya Year 1 composite score has a 50-point maximum, and the upper tail of both baseline and endline distributions reaches that ceiling. Score compression could attenuate treatment effects for the highest-ability students, but this would work against finding upper-track harm and against finding a positive treatment effect, so it makes the Kenya null conservative. A related concern is that tracking might compress or expand the overall distribution of scores. Table ?? shows that it does not: tracking redistributes variance between and within classrooms without changing the aggregate variance, so the null average effect in Kenya is not an artifact of distributional compression.

7 Conclusion

The general problem this paper addresses extends beyond ability grouping. Many policy settings involve replacing a coarse but established assignment rule with a finer signal: algorithmic risk scores

replace judicial discretion, standardized assessments replace teacher judgment, credit models replace relationship banking. The appeal is always the same—the new signal contains more information about the relevant type. Our experiments show that the information content of the new signal is only half the question. The other half is what happens to the production environment when you act on it.

In our setting, grade is the incumbent rule. It is coarse on measured achievement, but it bundles students by age, developmental stage, curricular exposure, and social cohort. A baseline assessment captures current performance more precisely, but acting on it unbundles these dimensions: it mixes age groups, disrupts peer composition, changes class sizes, and reorganizes the classroom as a production unit. In Liberia, where the baseline explains less than 7% of endline variation, this unbundling yields almost no informational gain while substantially worsening the classroom environment. In Kenya, where the baseline explains over 50%, regrouping does improve instructional match—yet mean achievement still does not rise, because the classroom costs of acting on the signal are large enough to offset the targeting gains.

This tension between informational precision and organizational disruption is not unique to schools. Any assignment system that replaces an incumbent rule with a finer one faces a version of it. The incumbent rule, however crude, is typically embedded in an organizational equilibrium: teachers know how to manage age-homogeneous classrooms, hospitals are staffed for standard patient flows, insurance pools reflect established risk-sharing arrangements. A signal that improves classification on one dimension may still worsen outcomes if it disrupts the organizational structure that made the old rule functional. The economics of finer sorting therefore cannot be evaluated from the signal alone. They depend jointly on how much the signal improves on the incumbent rule and what the act of reorganizing around it does to the production process.

The evidentiary basis differs across claims. The average treatment effects are experimental, identified by school-level randomization. The diagnostic-quality contrast and the finding that Liberia’s sorting fails mechanically are direct empirical observations requiring no structural assumptions. The peer-effects estimates in Kenya are suggestive—they exploit within-school variation under maintained assumptions that we probe but cannot definitively verify. The structural decomposition provides a qualitative organizing lens rather than a precise quantitative test.

The practical implication for education systems is that the bar for adopting ability grouping

should include not only evidence that a diagnostic predicts achievement, but evidence that the diagnostic improves on grade by enough to survive classroom reorganization. Neither of our experiments clears that bar. In Liberia, the signal is too weak for the question to be close. In Kenya, the signal is strong and the answer is still ambiguous.

Moreover, signal quality is not an exogenous parameter that implementers can freely choose. It is endogenous to the institutional environment: the same conditions that generate wide within-grade dispersion—less standardized instruction, less trained teachers, wider heterogeneity in prior exposure—also make a one-time diagnostic less predictive. The demand for ability grouping is therefore highest where its feasibility is lowest. And as Remark 1 formalizes, cross-grade grouping faces a structurally harder version of this problem than within-grade tracking: it requires the diagnostic to produce commensurable rankings across grades and imposes organizational costs—grade mixing, class-size reallocation—that within-grade tracking avoids by construction. The positive effects found by Duflo et al. (2011) for within-grade tracking are consistent with this distinction. The scope for cross-grade grouping to raise achievement is likely narrower, and the more promising path may be interventions that improve instructional targeting within existing classroom structures rather than interventions that reorganize classrooms around a new signal.

More broadly, these findings point to a gap in how targeting interventions are typically evaluated. The standard approach asks whether a signal predicts the outcome of interest. That is necessary but not sufficient. The question is whether the signal improves on the incumbent rule by enough to compensate for the costs of reorganizing the production environment around it. In settings where production is collective—classrooms, teams, hospital wards—acting on a better signal always has organizational consequences, and those consequences are part of the policy’s cost. A framework that evaluates signals in isolation will systematically overstate the returns to finer sorting.

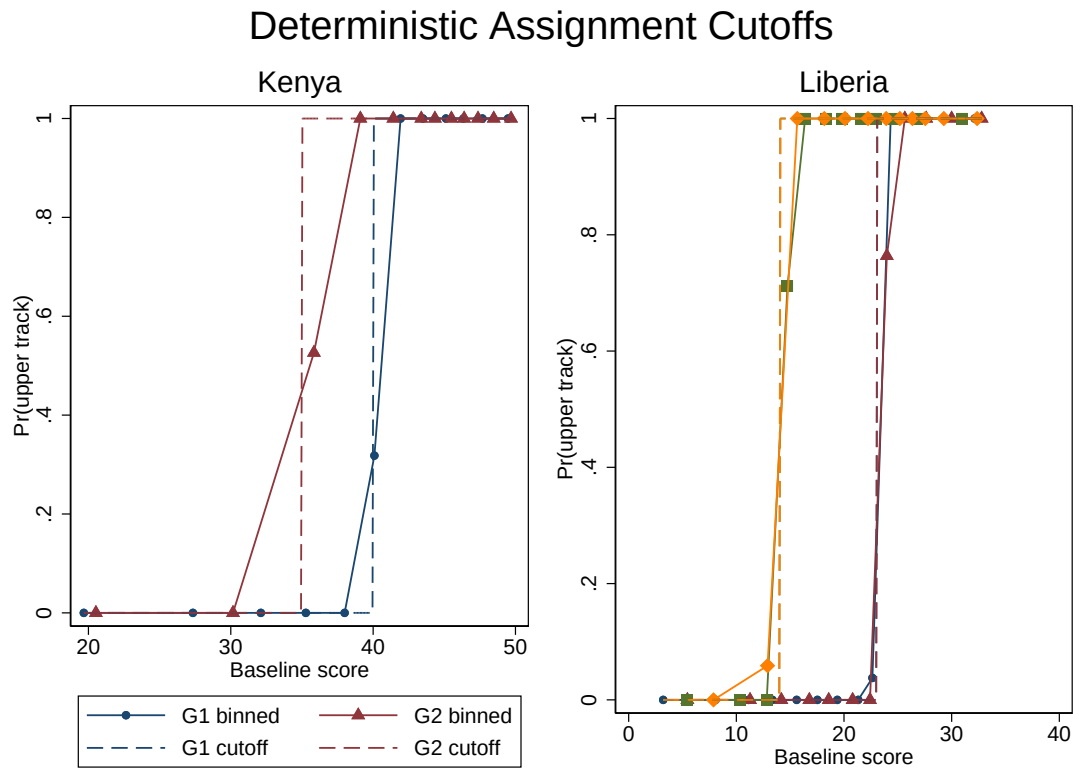
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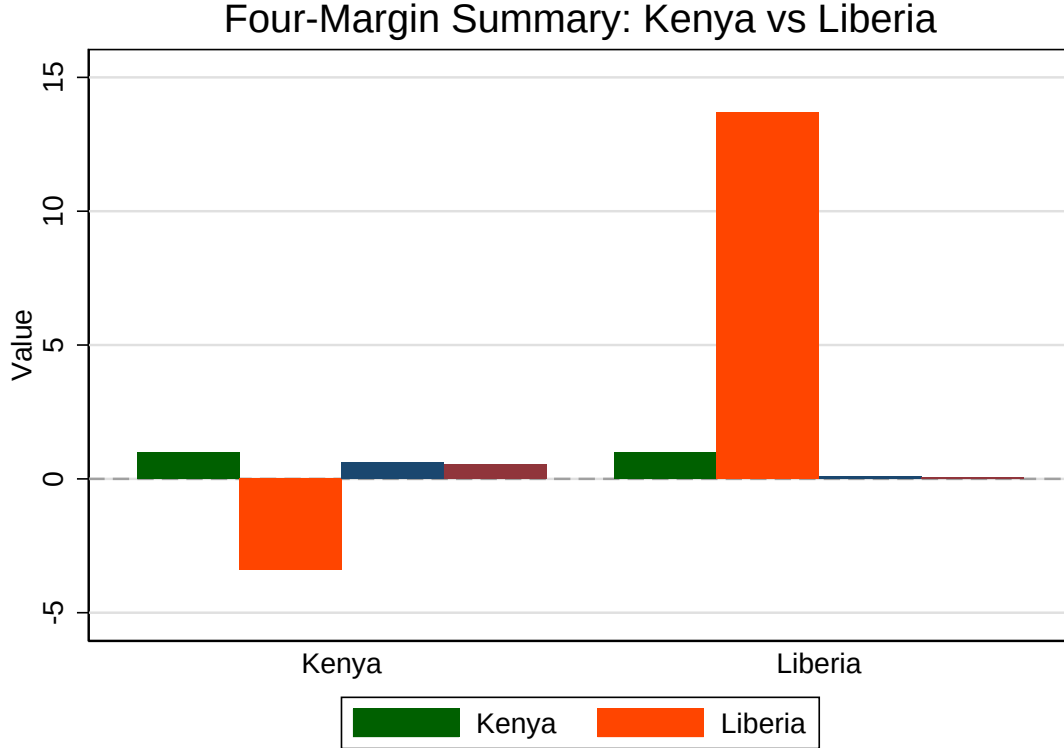
Figures

Figure 7: Deterministic Assignment Cutoffs



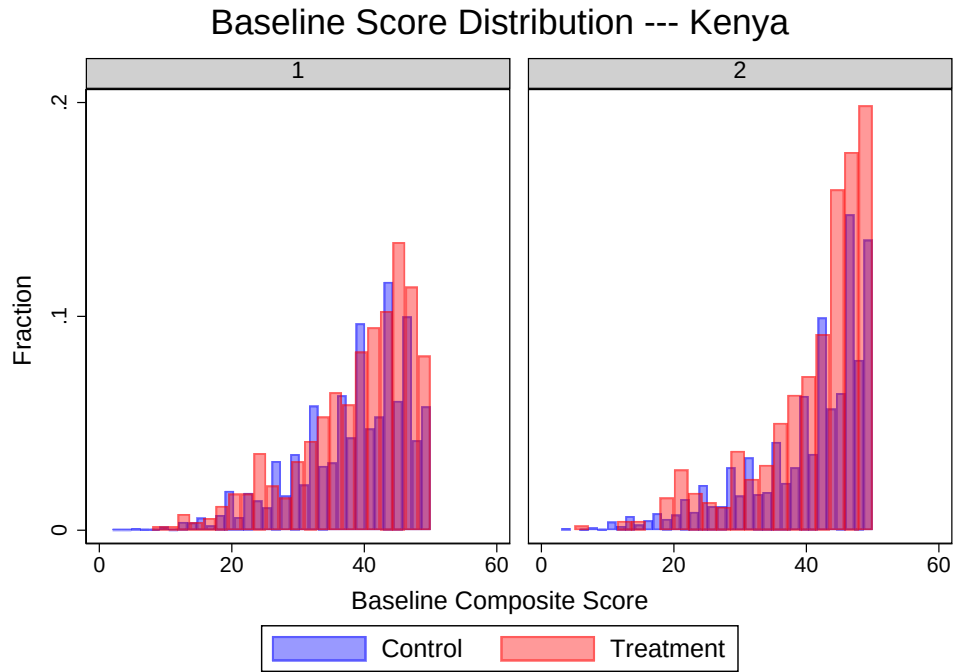
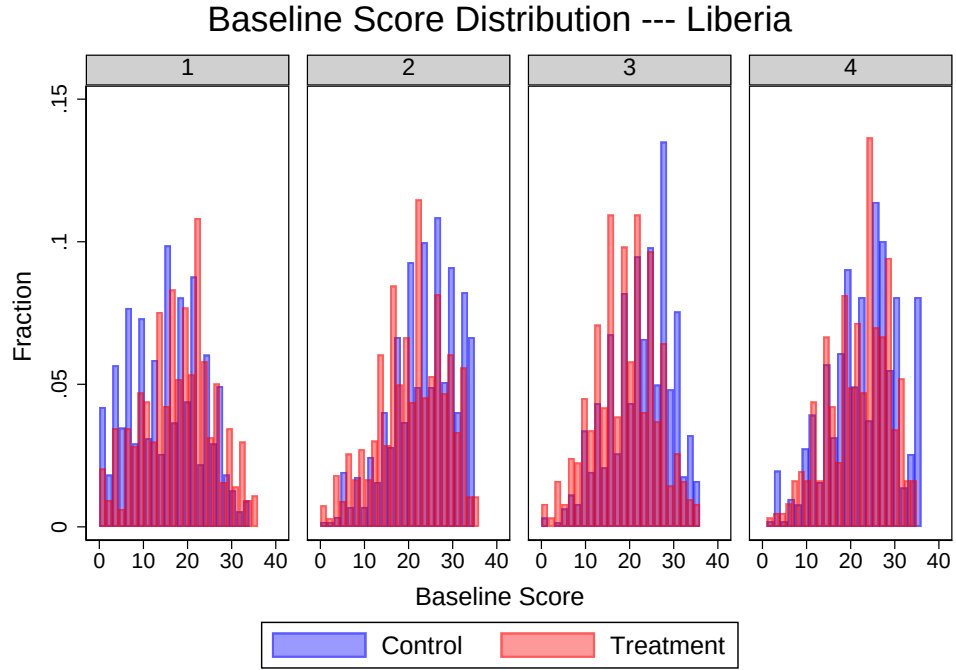
Notes: This figure plots the empirical probability of assignment to the upper reading track against the raw baseline score for students in treated schools. Panels correspond to the grade-specific assignment rules in Kenya Year 1 and Liberia. The discontinuous jumps occur exactly at the published score thresholds, showing that schools implemented deterministic cutoffs with no observed assignment error in the analytic sample.

Figure 1: Kenya-Liberia Contrast on Four Assignment Margins



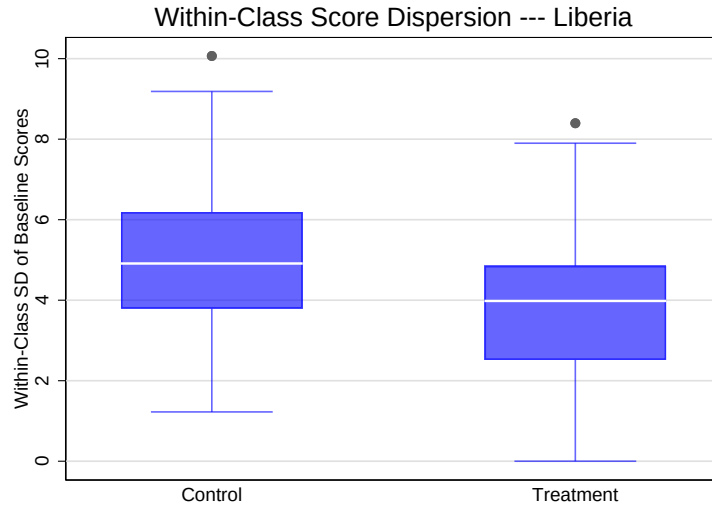
Notes: This summary figure compares Kenya Year 1 and Liberia along the four margins that organize the paper. Panel (a) reports weighted baseline predictive content (R_g^2), Panel (b) reports assignment sharpness (one minus misclassification rate), Panel (c) reports track-target mismatch reduction (control minus treatment means), and Panel (d) reports class-size reallocation (treatment minus control means). Both settings implement deterministic assignment and reduce mismatch, but Kenya has much stronger predictive content and smaller treatment classes, while Liberia has weak predictive content and much larger treatment classes.

Figure 2: Baseline Score Distributions

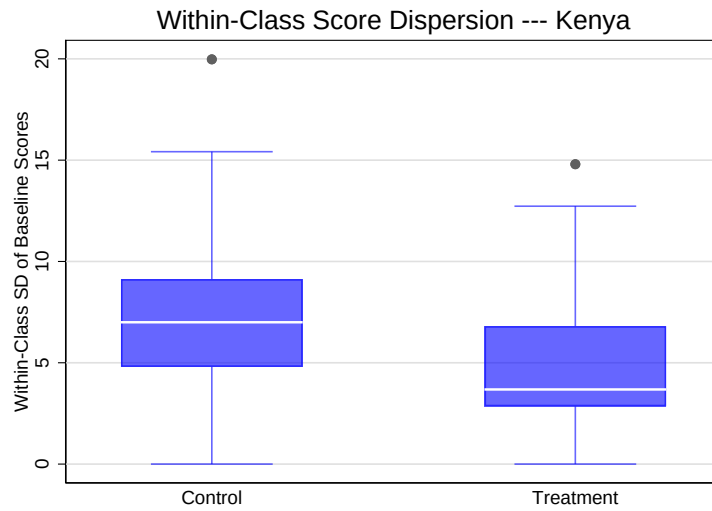


Notes: This figure plots kernel density estimates of raw baseline reading scores for students in the analytic sample, separately by study, enrolled grade, and treatment assignment. The vertical dashed lines mark the score cutoffs used to assign students to the upper reading track in treated schools. The substantial overlap across adjacent grades shows why replacing grade with a finer score-based assignment rule is feasible in both settings.

Figure 3: Within-Class Dispersion: Treatment vs. Control



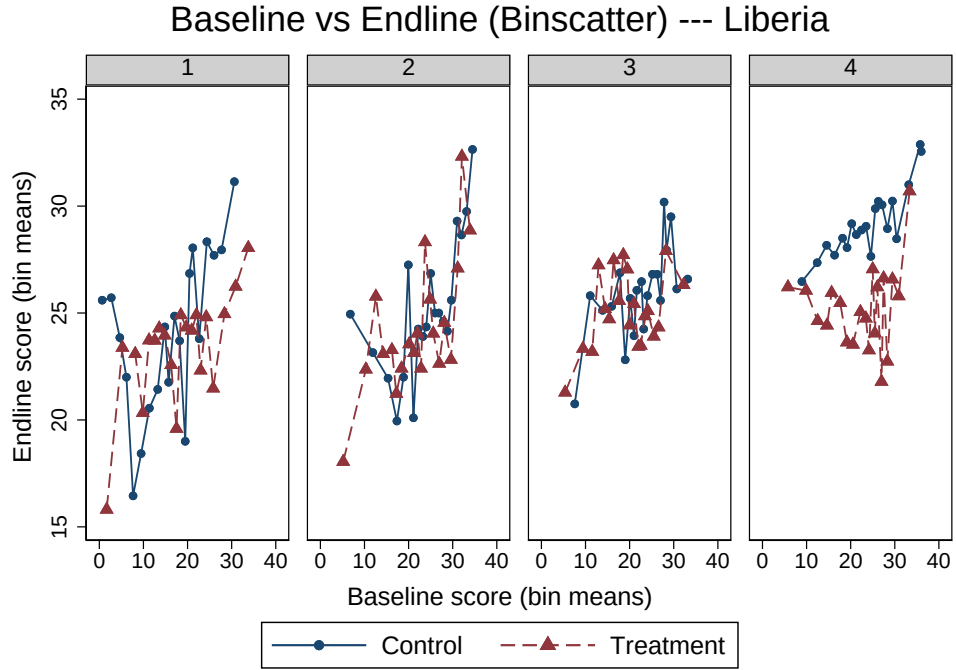
(a) Liberia



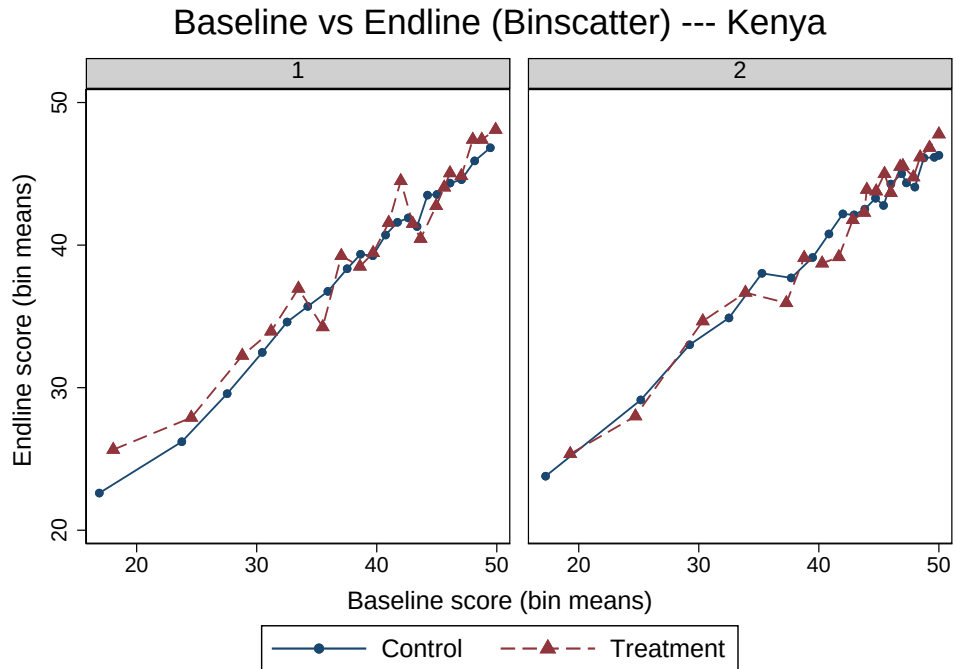
(b) Kenya Year 1

Notes: Each panel shows the distribution of the classroom-level standard deviation of baseline reading scores for realized reading classrooms in treatment and control schools. The unit of observation is a reading classroom. A leftward shift indicates greater within-class homogeneity. The compression is stronger in Kenya Year 1 than in Liberia.

Figure 4: Baseline vs. Endline Scores



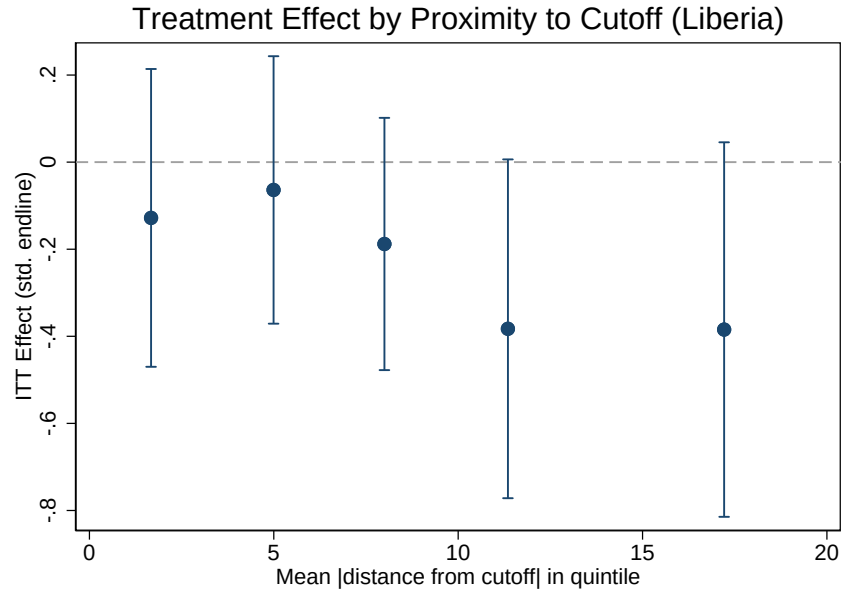
(a) Liberia



(b) Kenya Year 1

Notes: This figure plots binned means of endline reading scores against baseline reading scores for students in the analytic sample, separately by study, enrolled grade, and treatment assignment. Higher slope indicates that baseline achievement is more informative about later achievement. The stronger baseline-endline comovement in Kenya Year 1 than in Liberia mirrors the difference in baseline predictive power reported in Table 8.

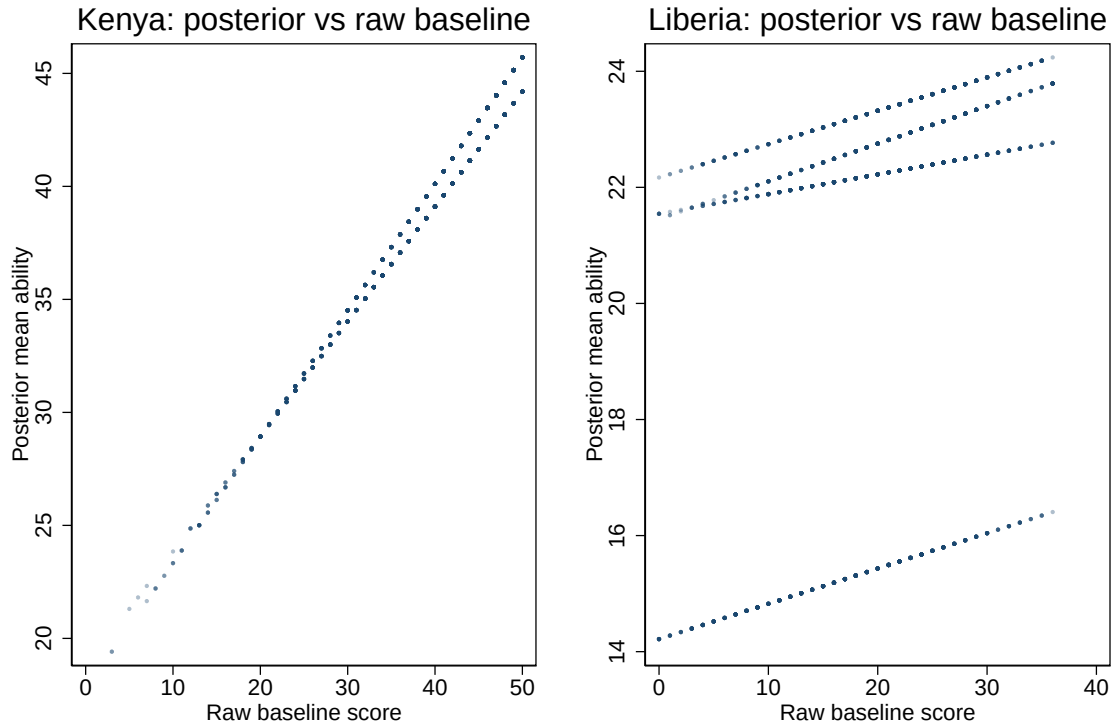
Figure 5: Treatment Effect by Distance from Assignment Cutoff in Liberia



Notes: Bars report Liberia intent-to-treat estimates for standardized endline reading scores, separately by quintiles of a student's absolute distance from the grade-specific assignment cutoff. The underlying estimates come from student-level OLS regressions with strata fixed effects and the empirical Bayes baseline control used elsewhere in the paper; error bars denote 95% confidence intervals based on cluster-robust standard errors. If harm came mainly from students being marginally misassigned near the threshold, the largest losses would appear in the leftmost bins. Instead, treatment effects become more negative farther from the cutoff.

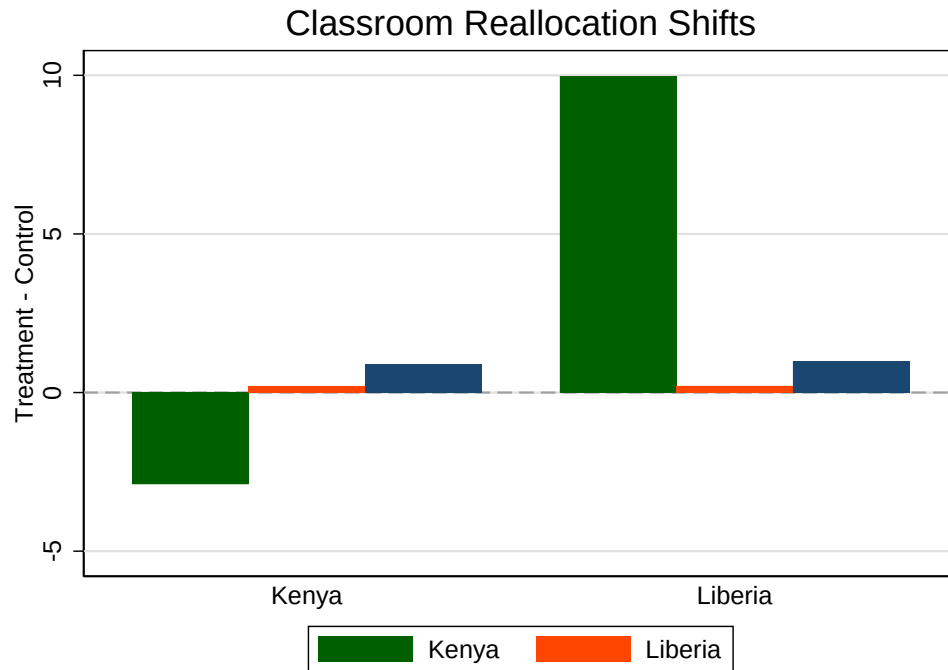
Figure 6: Posterior Shrinkage and Signal Quality

Posterior Shrinkage Comparison



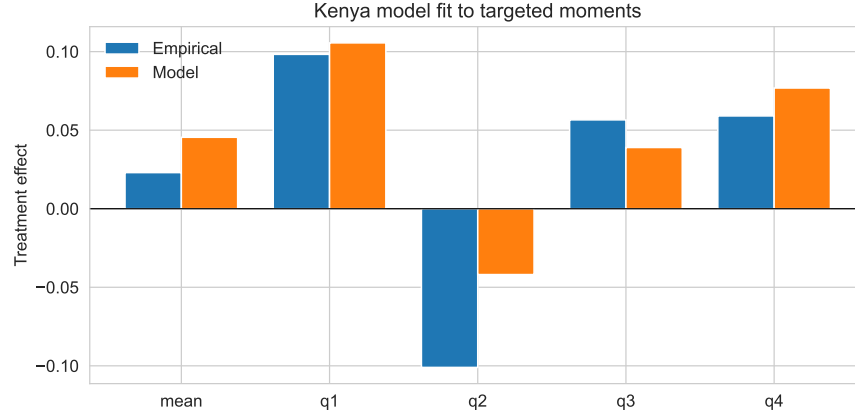
Notes: The top panels show the distribution of raw baseline reading scores by grade in each study. The bottom panels map the empirical Bayes posterior mean of latent ability to the raw baseline score using the grade-specific baseline predictive power estimates in Table 8. When signal quality is low, the posterior is pulled strongly toward the grade mean; when signal quality is high, the posterior tracks the raw score more closely. Liberia therefore exhibits much stronger shrinkage than Kenya Year 1.

Figure 8: Treatment-Induced Classroom Reallocation



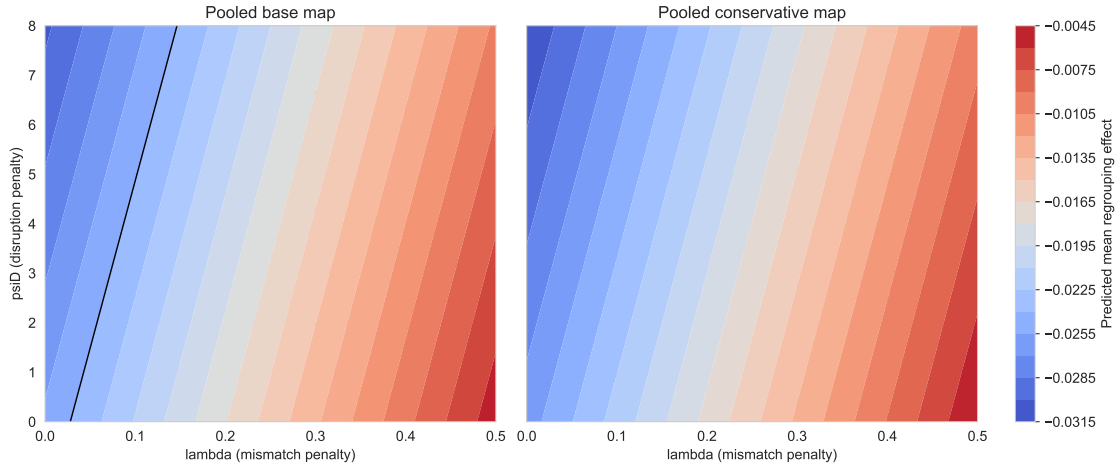
Notes: Bars show treatment minus control differences in classroom objects measured from realized reading classrooms in the structural analysis sample. The three outcomes are mean class size, mean within-class variance of enrolled grade, and mean number of grades represented in the reading classroom. Positive values indicate that treatment increases the object. Cross-grade grouping reduces class size in Kenya Year 1, increases class size in Liberia, and increases grade mixing in both settings.

Figure 9: Kenya Year 1 Model Fit to Targeted Moments



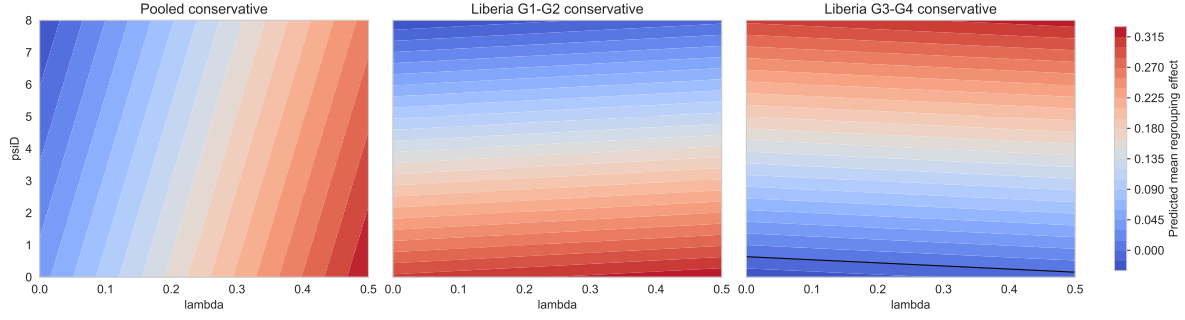
Notes: This figure compares observed Kenya Year 1 treatment effects in standardized endline reading achievement with the model-implied moments from the preferred Kenya-only minimum-distance estimator. The plotted targets are the overall intent-to-treat effect and the intent-to-treat effects by baseline quartile. Close alignment indicates that the estimator reproduces the moments it is designed to match.

Figure 10: Liberia Sensitivity of Predicted Regrouping Gains



Notes: The figure reports pooled Liberia sensitivity maps over a grid of the mismatch penalty λ and the classroom-disruption penalty ψ_D . The left panel uses the baseline specification for the Liberia policy-contrast exercise. The right panel adds conservative assumptions on the social-composition term and fixed regrouping costs. Colors show the model-implied mean effect of regrouping on endline achievement, with warmer colors indicating more favorable predicted effects.

Figure 11: Liberia Sensitivity of Predicted Regrouping Gains by Grade Block



Notes: These conservative sensitivity maps repeat the Liberia policy-contrast exercise for the pooled sample, the Grades 1–2 block, and the Grades 3–4 block. Each panel plots the predicted mean regrouping effect over the same grid of mismatch and classroom-disruption penalties. The figure shows that the pooled Liberia result masks substantial heterogeneity across the two grade blocks, with a much less favorable region for Grades 1–2 than for Grades 3–4.

Tables

Table 1: Summary Statistics — Liberia

Variable	Control Mean	Treatment Mean	SD (All)	N
Baseline score	21.014	19.848	7.936	4784
Endline score	25.983	24.467	7.612	3154
Upper-track assignment	0.612	0.549	0.494	4784
Has endline	0.659	0.660	0.474	4784
Class size	45.596	59.278	27.288	4784

Notes: Means and standard deviations are reported in the analytic sample (fin-samp==1). Endline outcomes are based on observations with non-missing endline test scores.

Table 2: Summary Statistics — Kenya

Variable	Control Mean	Treatment Mean	SD (All)	N
Baseline score	39.207	39.904	8.786	7102
Endline score	39.333	40.258	8.519	4954
Upper-track assignment	0.614	0.652	0.486	7102
Has endline	0.686	0.771	0.459	7102
Class size	24.307	20.935	11.233	7102

Notes: Means and standard deviations are reported in the analytic sample (fin-samp==1). Endline outcomes are based on observations with non-missing endline test scores.

Table 3: Baseline Balance — Liberia

	Control Mean	Coefficient	N
Baseline Score	21.014	-1.136 (1.017)	4784
Upper Group	0.612	-0.056 (0.046)	4784
Has Endline	0.659	-0.000 (0.062)	4784
Class Size	45.596	13.484*** (4.482)	4784

Notes: All specifications control for randomization strata fixed effects. Standard errors are clustered at the school-grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 4: Baseline Balance — Kenya

	Control Mean	Coefficient	N
Baseline Score	39.207	0.860 (0.661)	7102
Upper Group	0.614	0.042 (0.034)	7102
Has Endline	0.686	0.061 (0.046)	7102
Class Size	24.307	-3.071* (1.831)	7102

Notes: All specifications control for randomization strata fixed effects. Standard errors are clustered at the school-grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 5: Effect of Ability Grouping on Test Scores

	Kenya	Liberia
Treatment	0.034 (0.057)	-0.212 (0.135)
Control Mean	0.012	0.091
N	4624	3154

Notes: Dependent variable is standardized endline score. All specifications control for EB ability and strata FE. Standard errors clustered at school (Kenya) or school-grade-group (Liberia) level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 13: Kenya Year 1 minimum-distance accounting estimates

(1)	
Mismatch penalty, $\hat{\lambda}$	0.0496
Classroom-disruption term, $\hat{\phi}$	0.3392
Social-composition term, $\hat{\zeta}$	-0.1047
Objective value	0.00105
Converged	Yes

Notes: The table reports the preferred Kenya Year 1 minimum-distance estimator from the accounting exercise. The estimator targets the Kenya moments emphasized in the text, including the mean treatment effect, treatment-effect heterogeneity by baseline quartile, and the treatment-induced changes in mismatch and classroom composition. $\hat{\lambda}$ is the mismatch penalty, $\hat{\phi}$ is the classroom-disruption term, and $\hat{\zeta}$ is the social-composition term. The objective value is the minimized weighted distance between the estimated and target moments.

Table 6: Treatment Effects by Track

	Kenya	Liberia
Lower track (Treatment)	0.052 (0.147)	-0.019 (0.181)
T $\times$ Upper	-0.026 (0.157)	-0.328 (0.201)
Upper track total	0.025. (0.045)	-0.348** (0.154)
N	4624	3154

Notes: Upper track total is Treatment + T $\times$ Upper. Controls: EB ability, strata FE. Cluster-robust SEs. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 7: Peer Composition Effects (Borusyak-Hull)

	Kenya	Liberia
Peer EB Ability	-0.125** (0.055)	-0.011 (0.079)
N	4624	3153

Notes: Dependent variable is standardized endline score. Peer EB ability is the leave-self-out mean of EB ability in the student's classroom. Controls: baseline-decile $\times$ treatment FE, strata FE. Cluster-robust SEs. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 8: Diagnostic Reliability: R^2 (Baseline $\rightarrow$ Endline)

Study	Grade	N (Control)	ρ	ρ^2
Kenya	1	2190	0.713	0.509
	2	2005	0.748	0.559
Liberia	1	411	0.247	0.061
	2	399	0.240	0.057
	3	324	0.184	0.034
	4	348	0.255	0.065

Notes: Pearson correlation between baseline and endline raw scores in the control group, by grade. ρ^2 measures the fraction of endline variance predicted by the baseline diagnostic.

Table 9: Alternative Signal-Quality Diagnostics

	Kenya	Liberia
R^2 grade FE only	0.011	0.059
R^2 grade FE + baseline	0.534	0.109
Incremental R^2	0.523	0.050
Within-grade rank persistence	0.684	0.242

Notes: All computed in control group. Incremental R^2 = additional variance explained by adding baseline score to a grade-FE-only model. Rank persistence is the within-grade Spearman correlation between baseline and endline score ranks, weighted by grade size.

Table 10: Recovered deterministic assignment cutoffs

Grade	Kenya Year 1			Liberia		
	Cutoff	N treated	Misclass. rate	Cutoff	N treated	Misclass. rate
1	40	527	0.000	23	637	0.000
2	35	458	0.000	23	662	0.000
3				14	621	0.000
4				14	615	0.000

Notes: The table reports score thresholds used to assign treated students to the upper reading track. Misclassification is the share where observed assignment differs from the deterministic cutoff rule.

Table 11: Classroom reallocation under cross-grade grouping

	Kenya Year 1		Liberia	
	Control	Treat	Control	Treat
Class size	20.88	17.99	32.19	42.16
Within-class grade variance	0.000	0.209	0.000	0.211
Number of grades in class	1.000	1.896	1.000	1.981
Track-target mismatch	1.352	0.493	1.519	0.816
Peer mean EB ability	-0.006	0.254	0.009	0.032

Notes: Entries are arm-specific means in the structural analysis sample. Realized classroom objects use treated-track classes in treatment and grade classes in control.

Table 12: Sufficient-Statistics Decomposition of Kenya ITT

	Preferred		Alternative	
	Estimate	SE	Estimate	SE
<i>Panel A: Marginal returns</i>				
Mismatch penalty ($\hat{\lambda}$)	-0.0311	(0.0228)	0.0921	(0.0665)
Peer/rank composite ($\hat{\zeta}$)	-0.1340**	(0.0556)	-0.1384	(0.0561)
<i>Panel B: Treatment-induced shifts</i>				
Delta —Mismatch— (C - T)	0.4273			
Delta Mismatch squared (C - T)	0.9146			
Delta Peer composition (T - C)	0.2955			
Delta Class size (T - C)	-2.986			
<i>Panel C: ITT decomposition (preferred)</i>				
Overall ITT	0.0342	(0.0566)		
Mismatch channel (lambda x Delta Misfit)	0.0133	(0.0096)		
Peer/rank channel (zeta x Delta Peer)	-0.0396	(0.0202)		
Residual	0.0605	(0.0565)		

Notes: Preferred mismatch specification uses $|\hat{\theta} - \hat{I}|$ with linear ability control in the full sample. Alternative mismatch specification uses $(\hat{\theta} - \hat{I})^2$ in treated schools with track FE. Preferred peer/rank specification uses decile $\times$ treatment $\times$ grade FE; alternative uses ventile cells. Panel C uses preferred estimates, and bootstrap SEs are clustered at school level.

Table 14: Liberia Sufficient Statistics and Transported Decomposition

	Estimate	SE
<i>Panel A: Peer estimate</i>		
Liberia BH peer estimate ($\hat{\zeta}_{Lib}$)	-0.0481	(0.1077)
Transported Kenya peer estimate ($\hat{\zeta}_{Ken}$)	-0.1340	(0.0556)
<i>Panel B: Liberia shifts</i>		
Delta —Mismatch— (C - T)	0.3482	
Delta Peer composition (T - C)	0.0411	
Delta Class size (T - C)	11.375	
<i>Panel C: Transported decomposition</i>		
Overall Liberia ITT	-0.2112	(0.1351)
Mismatch channel (lambda Ken x Delta Misfit)	0.0108	
Peer channel (zeta Ken x Delta Peer)	-0.0055	
Residual	-0.2166	

Notes: Liberia peer estimates use BH conditioning with decile $\times$ treatment $\times$ grade FE and strata FE. Panel C transports the preferred Kenya estimates into Liberia shifts; residual is the remainder needed to match the Liberia ITT.

Table 15: Liberia Sensitivity: Predicted Effect under Kenya Parameters

λ	Disruption cost (ψ)				
	0.0	0.1	0.3	0.5	1.0
0.00	-0.006	-1.143	-3.418	-5.693	-11.380
0.05	-0.023	-1.160	-3.435	-5.710	-11.398
0.10	-0.040	-1.178	-3.453	-5.728	-11.415
0.20	-0.075	-1.213	-3.488	-5.763	-11.450
0.30	-0.110	-1.247	-3.522	-5.797	-11.485
0.50	-0.180	-1.317	-3.592	-5.867	-11.554

Notes: Each cell uses Liberia treatment-control shifts with predicted effect = $\lambda \times (\text{mean mismatch in treatment minus control}) + \text{zeta Ken} \times \text{Delta Peer} - \psi \times \text{Delta Class size}$. Mismatch is defined as absolute deviation from track target.

Table 16: ITT Heterogeneity by Distance from Assignment Cutoff (Liberia)

Quintile	Mean $ s - c $	ITT	SE	N
Q1	1.7	-0.128	(0.174)	813
Q2	5.0	-0.064	(0.157)	636
Q3	8.0	-0.188	(0.148)	556
Q4	11.3	-0.383*	(0.198)	604
Q5	17.2	-0.385*	(0.219)	545
$T \times s - c $		-0.0253	(0.0176)	3154

Notes: Quintiles of absolute distance from grade-specific assignment cutoff. OLS with strata FE, EB ability control, cluster-robust SEs.

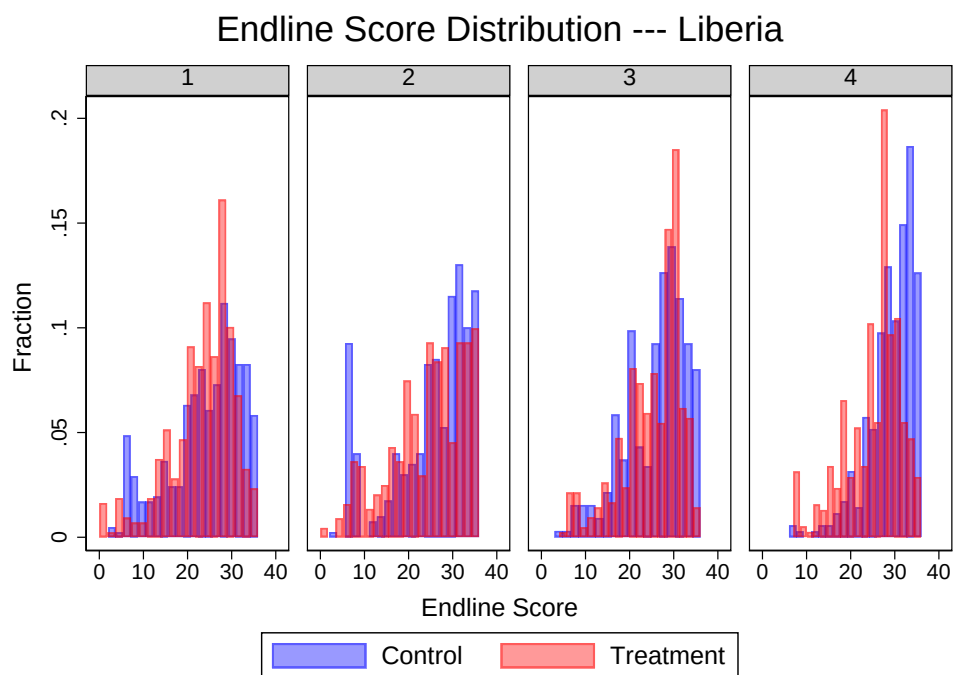
Table 17: Treatment Effects by Track and Ability Tercile (Liberia)

Track	Ability Bin	ITT	SE	p	N
Lower	Bottom	-0.192	(0.214)	0.372	460
	Middle	0.150	(0.283)	0.599	459
	Top	-0.055	(0.238)	0.819	393
Upper	Bottom	-0.204	(0.174)	0.244	626
	Middle	-0.472*	(0.242)	0.055	606
	Top	-0.354**	(0.150)	0.021	610

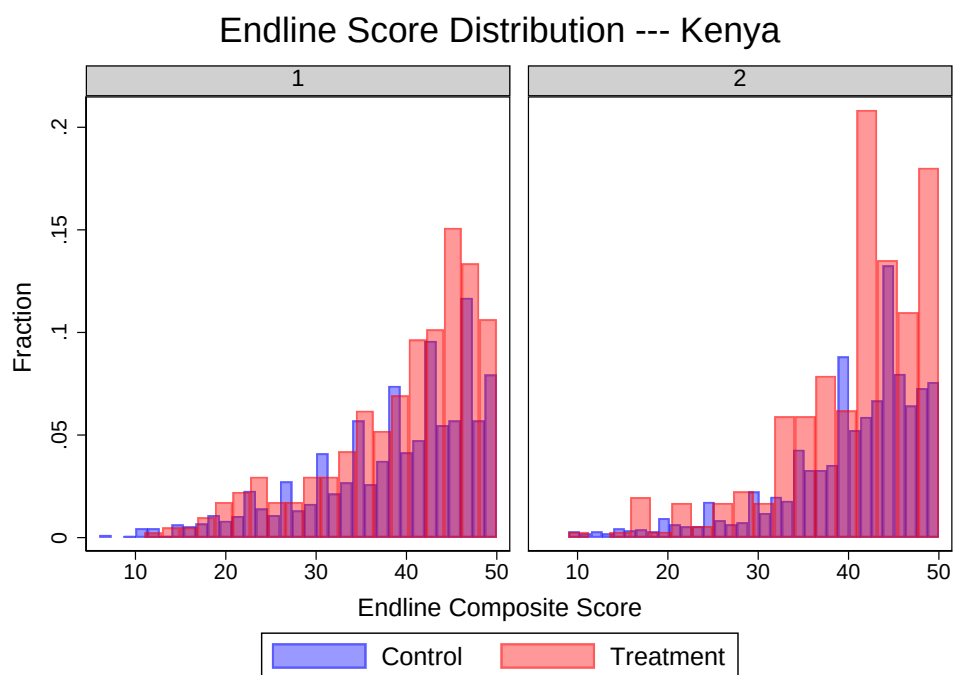
Notes: Within-track ability terciles. OLS with strata FE, EB control, cluster-robust SEs. ***, **, and * indicate significance at 1%, 5%, and 10%.

A Supplementary Descriptives and Diagnostics

Figure A1: Endline Score Distributions by Grade and Treatment Status

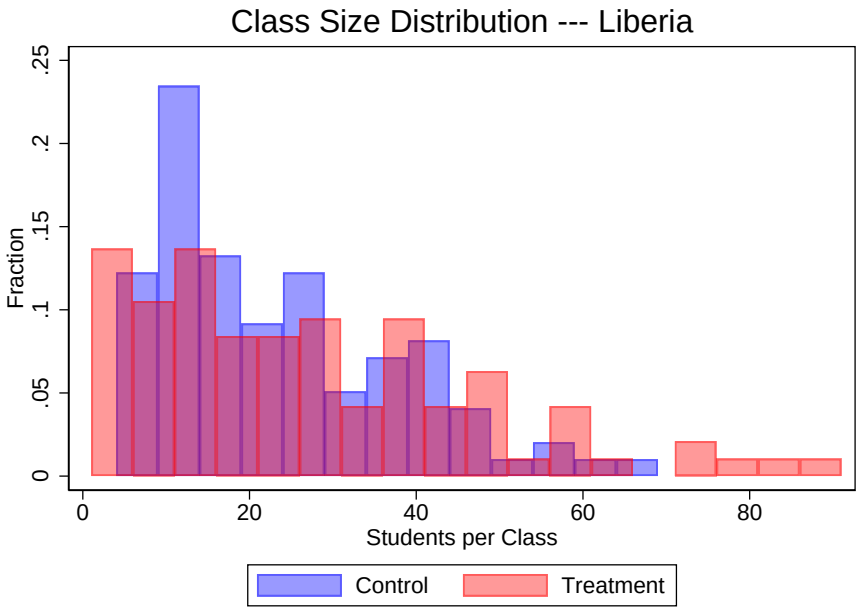


(a) Liberia

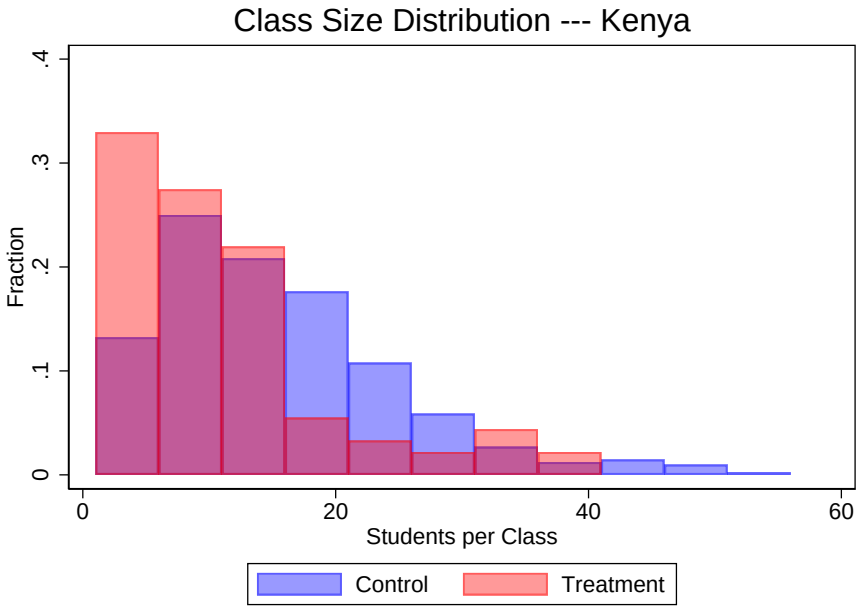


(b) Kenya Year 1

Figure A2: Reading-Class Size Distributions by Treatment Status



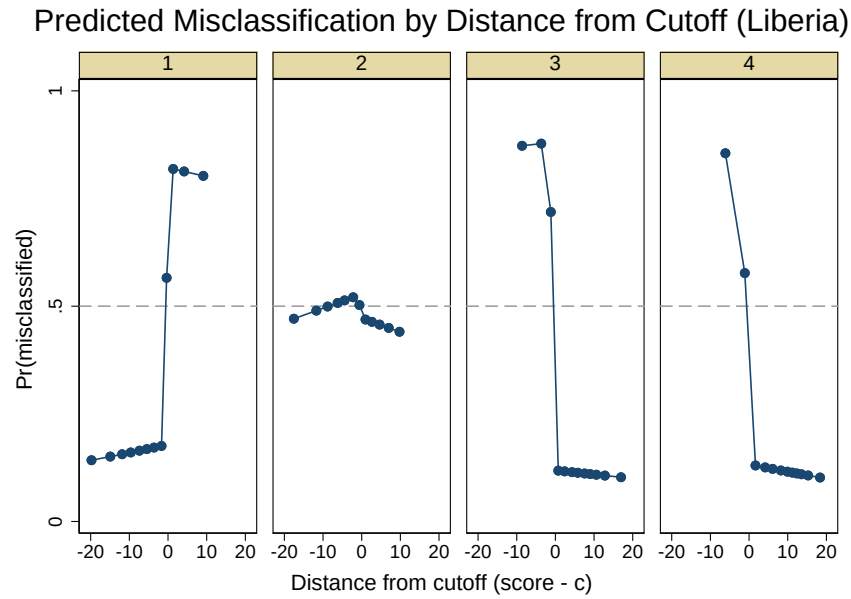
(a) Liberia



(b) Kenya Year 1

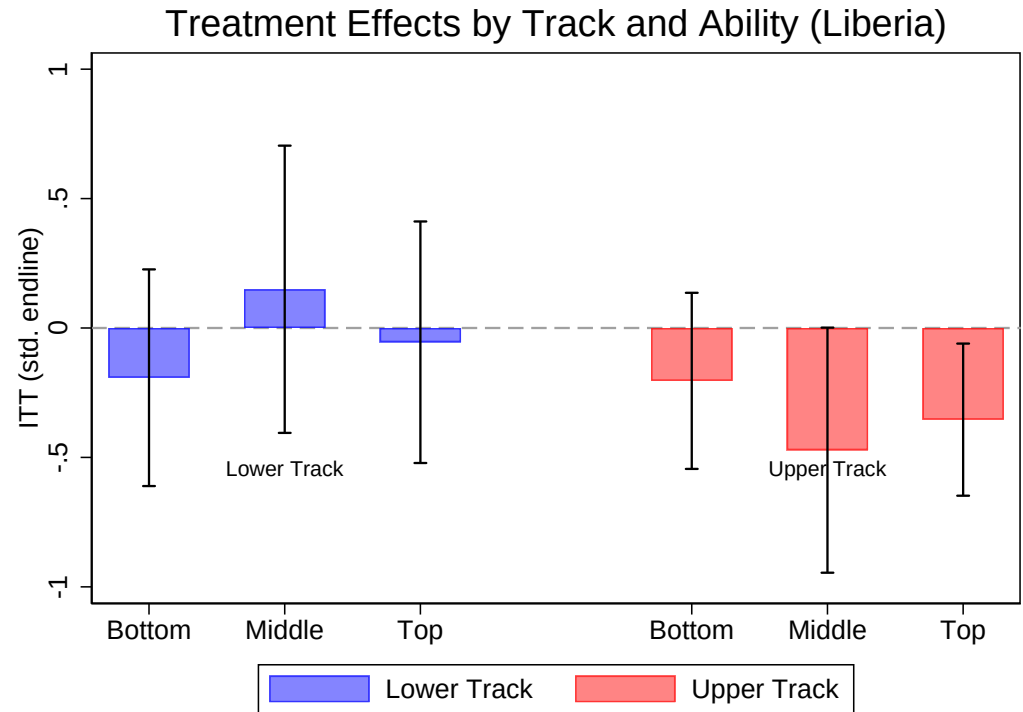
Notes: Each panel shows the distribution of realized reading-class size, measured as the number of students assigned to a classroom, in treatment and control schools. The unit of observation is a reading classroom. Control schools retain grade-based reading classrooms, while treatment schools reorganize students into cross-grade reading tracks. Regrouping substantially enlarges classrooms in Liberia but not in Kenya Year 1.

Figure A3: Predicted Misclassification by Distance from the Assignment Cutoff in Liberia



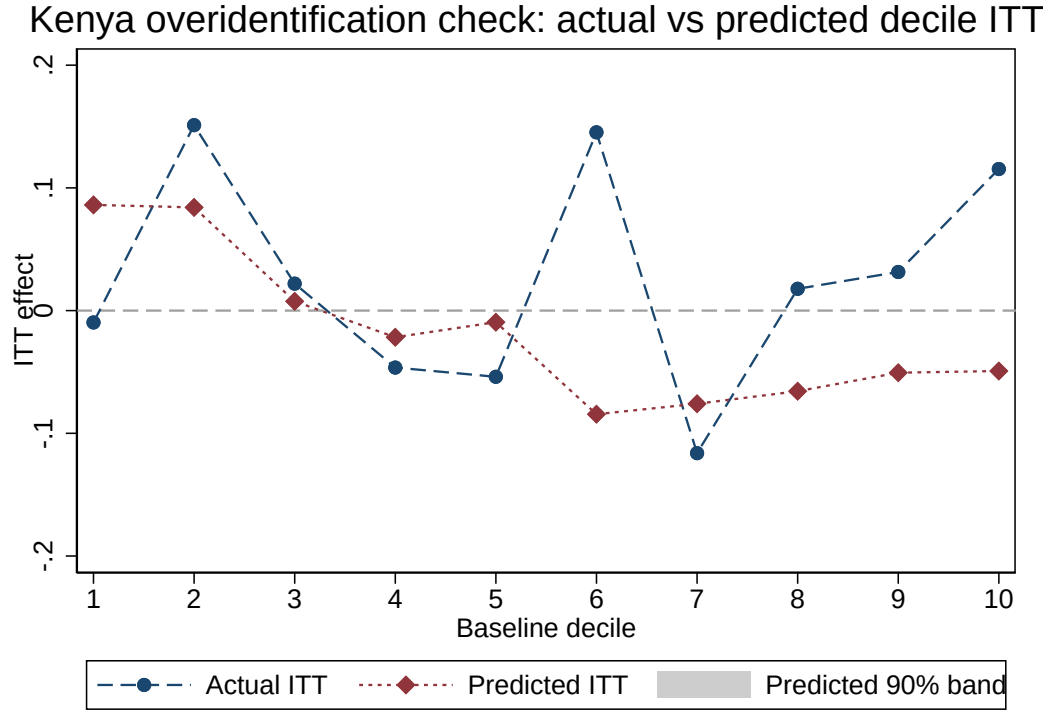
Notes: This figure plots the model-implied probability that a student's observed track assignment differs from the assignment that would obtain under latent ability, as a function of absolute distance from the Liberia grade-specific score cutoff. The probabilities are constructed from the Liberia empirical Bayes measurement system rather than from observed assignment errors, which are zero in the data. Under Liberia's low baseline predictive power, predicted misclassification remains substantial even for students away from the cutoff.

Figure A4: Liberia Treatment Effects by Track and Within-Track Ability Tercile



Notes: Bars report Liberia student-level intent-to-treat estimates for standardized endline reading scores, separately by assigned track and by tercile of baseline ability within that track. Error bars denote 95% confidence intervals from the corresponding OLS specifications with strata fixed effects and the empirical Bayes baseline control. Red bars indicate estimates significant at the 5 percent level. The figure shows that the largest losses occur in the middle and top terciles of the upper track rather than among students nearest the lower-track margin.

Figure A5: Kenya Overidentification Check: Actual vs Predicted Decile ITTs



Notes: The figure compares decile-specific treatment effects estimated directly in the data (actual) with treatment effects predicted by the preferred sufficient-statistics decomposition. Shaded bands are bootstrap 90% confidence intervals for the predicted series.

Table A1: Kenya Overidentification Diagnostics

Statistic	Value
Correlation (actual vs predicted decile ITTs)	0.1274
RMSE of decile moments	0.1057
MAE of decile moments	0.0847

Notes: Predicted decile ITTs use preferred Kenya sufficient-statistics estimates (lambda from absolute mismatch and zeta from BH with decile-by-treatment-by-grade FE). Bootstrap 90% intervals are reported in the overidentification figure for predicted moments.

B Sample Flow

Table A2: Sample Flow — Liberia

Sample flow step	Control	Treatment	Total
All observations	3420	4165	7585
Analytic sample (finsamp=1)	2249	2535	4784
With baseline score	2249	2535	4784
With endline score	1482	1672	3154

Notes: Counts shown by treatment status at each sample-construction step in the cleaned analysis dataset.

Table A3: Sample Flow — Kenya

Sample flow step	Control	Treatment	Total
All observations	8271	1447	9718
Analytic sample (finsamp=1)	6117	985	7102
With baseline score	6117	985	7102
With endline score	4195	759	4954

Notes: Counts shown by treatment status at each sample-construction step in the cleaned analysis dataset.